

Affective Computing

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Chapter 1

Introduction

How do we define effective computing? computing that relates to, arises from, or influences emotions. It's a general definition that may comprise everything. Computing that is in relation with "emotion".

We have 3 task:

- recognise emotion: if i detect smiling face i can say that you're happy. Most of research is devoted to that.
- express emotion: we are assuming that any agent (robot, virtual) acting in some way (moving or virtual reality) expressing some emotion that we have to recognize. It's important by marketing point of view, but it's more computer graphics.
- have emotion: do this machine has emotion? this is not a simple question, affect is a key component of consciousness, there is no obvious answer of this.

1.1 Recognize emotion

We have some kind of scheme that is a variation of this. We might have the person that we are monitoring with body state (can we have emotion without a body or with a different body?). Nowadays we can record concrete physiological aspect. Also we can detect speech, social signal. Once we have collected physiological and behavioural variables they go in the detection part. We can also inject context, mixing all together in order to understand the internal state of human.

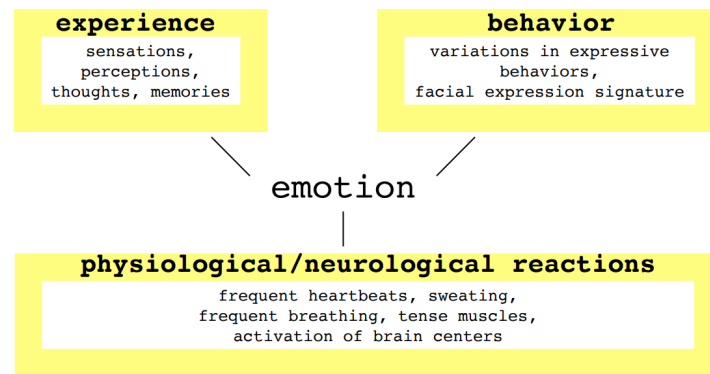
Here comes the problem, computer vision is very different, when we say recognition we talking about classifier, so how does computer vision perform this recognition? taking millions of images, label them and fill in an end-to-end machine that produce an output, compute the output with the label and propagate the error to adjust the labels. With normal object is easy label them, but how can we do this with emotions? in some case labelling emotion (means label what people has inside) we may do that in a certain circumstances, but more important aspect that we must have a common cultural background.

The story is that there is not certainly what is a label, so be carefully.

1.2 Emotion

What is an emotion? it's a kind of multidimensional phenomenon involving external action and reaction and internal action.

- Experience: stimulus, not strictly a physical stimulus for example memories
- Behavior: variations in expressive, facial expression
- Physiological/neurological reactions: you might not have emotion without a body experience. Also this has practical consequences because a lot of measurements concert this.



The first definition: "an emotion is a coordination or composite of experiences, behavioral expressions and physiological/neurological components with varying duration".

The methodology that'll follow is similar to the one of natural interaction. We have a context stimulus (context + event = people react), obviously we have to adapt our naive theory of emotion. We have two levels:

- Psychology: there is no clear distinction between cognitive and emotion. That means that we might imagine behaviour
- Neuroscience: when I go to neuroscience which is the right level of description? single neuron recording? no way, there is never a one to one from a one side of the brain to a point in the behavioural space. We have the idea that we have specialized areas in the brain, but actually that's not the full story, there are areas which are more connected with other areas but even in the case of the most segregated part of the visual cortex we might still have neurons which somehow participate in different tasks. It's very interesting group certain areas in network. The main problem is that there is not a one to one mapping but always a many to many mapping.

We are going to model this problem with PGM, probabilistic graphical model.

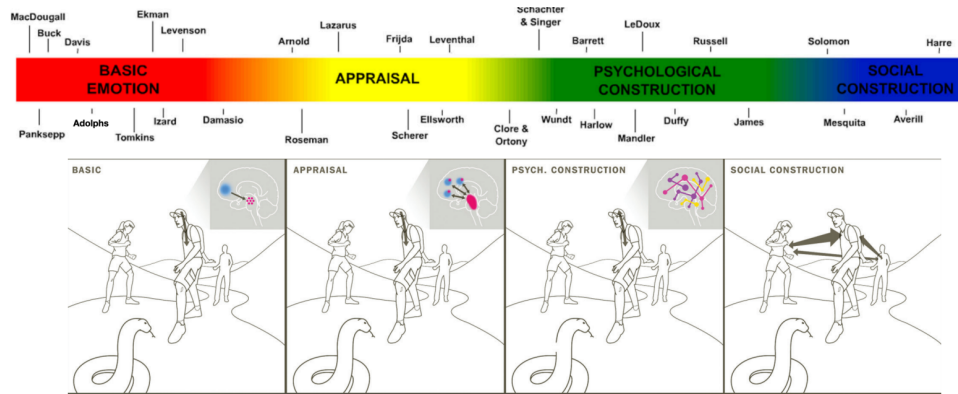
1.2.1 Timeline

First theory of emotion was by Plato and Aristotle, we nowadays apply these ideas: emotion is something disruptive and our reason must hold emotion and desire. The first how treated a theory of emotion was Descartes his idea was that emotion informs you about soul changes. Spinoza has become central after the work of Antonio Damasio (neuroscientist involved in theory of emotion), all starts with the basic motivation (conatus) that maintains us alive. Then the first scientific effort to define emotion was given by Darwin, set up emotion as evolutionary product. From that point studies involved both psychological and neuroscientific aspects.

Let's see how specifically studies deal with the fact that we receive a stimulus and we react to that. This is the spectrum, divided in 4 groups emotion theory:

- Basic emotion: inside out. Darwin interpretation.
- Appraisal: scientific version of Aristotle, emotions are result of evaluation of stimulus

- Psychological Construction: emotions are concepts and they are constructed. Much more complicated by the point of view of computational cost.
- Social Construction: it's a kind of extreme psychological construction theory that involve also social interaction. Emotion is kind of contract that we have with other.

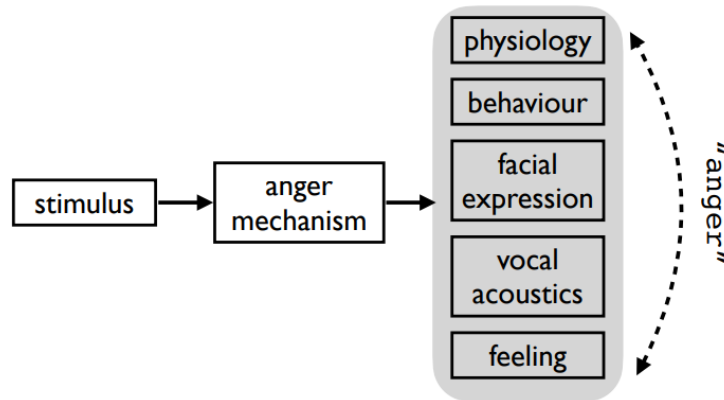


How do we exactly distinguish the theories? There are some formulated question that we want to answer:

					
Fundamental questions	Basic	Appraisal	Psychological Construction	Social Construction	
Are emotions unique mental states?	Yes	Yes	No	Varies by model	
Are emotions caused by special mechanisms?	Yes	Varies by model	No	No	
Is each emotion caused by a specific brain circuit?	Yes	No	No	No	
Do emotions have unique manifestations (in face, voice, body state)?	Yes	Varies by model	No	No	
Does each emotion have a unique response tendency?	Yes	In most models	No	No	
Is experience a necessary feature of emotion?	Varies by model	Yes	Yes	No	
What is universal?	Emotions are universal	Appraisals are universal	Psychological ingredients are universal	Influence of social context is universal	
How important is variability in emotions?	Epiphenomenal	Varies by model	Emphasized	Present, but not central	
Are emotions shared with non-human animals?	Yes	Some appraisals are shared	Affect is shared	No	
How did the evolution shape emotions?	Specific emotions evolved	Cognitive appraisals evolved	Basic ingredients evolved	Cultural and social structure evolved	

1.2.2 Basic Emotion Theory BET

Summary: emotion is triggered by a stimulus, each emotion (e.g. anger) having its own innate mechanisms.



This unique pattern identifies the emotion which we call anger (fear, happiness...).

Darwin was confronting animals and humans to find similar behaviour across the evolutionary.

We can see the our same 6 emotions also in animals like for example in apes.

According to BET we can define some general characteristic of non verbal expression of emotion:

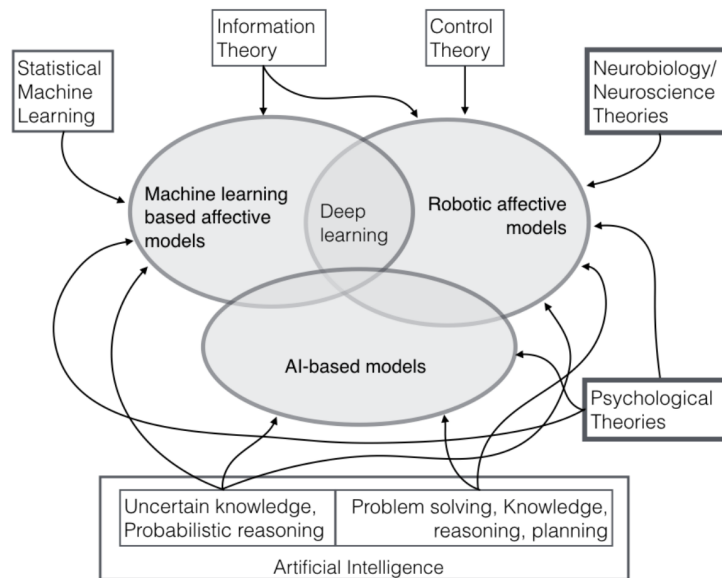
- they have brief, coherent patterns of behavior that tend to covary with distinct subjective experiences
- They signal the current emotional state, intentions, and/or assessment of the eliciting situation of the individual
- they manifest some degree of cross-cultural similarity in both production and recognition
- they find evolutionary precursors in the behaviors of other mammals in context similar to the contexts humans encounter
- they tend to covary with emotion-related physiological responses

Chapter 2

Emotion Theory

We have three areas:

- Machine Learning: this basically consider very simple models in emotions theory
- More general AI-based models: there are mostly adopted in some kind hybrid area
- Robotics: a robot is a very constrained environment

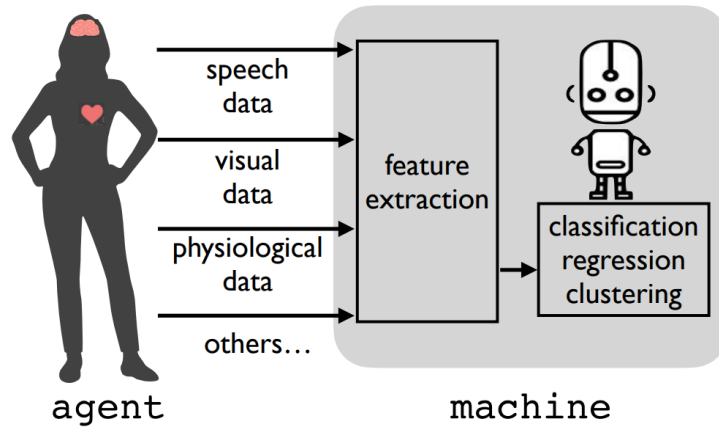


In this machine learning business the main players are statistical machine learning and it's recent evolution in deep learning, information theory. This provides one approach which is at least up to recently dealing with very simple emotion theory, but many of there paper don't even start from emotion theory but give from grant.

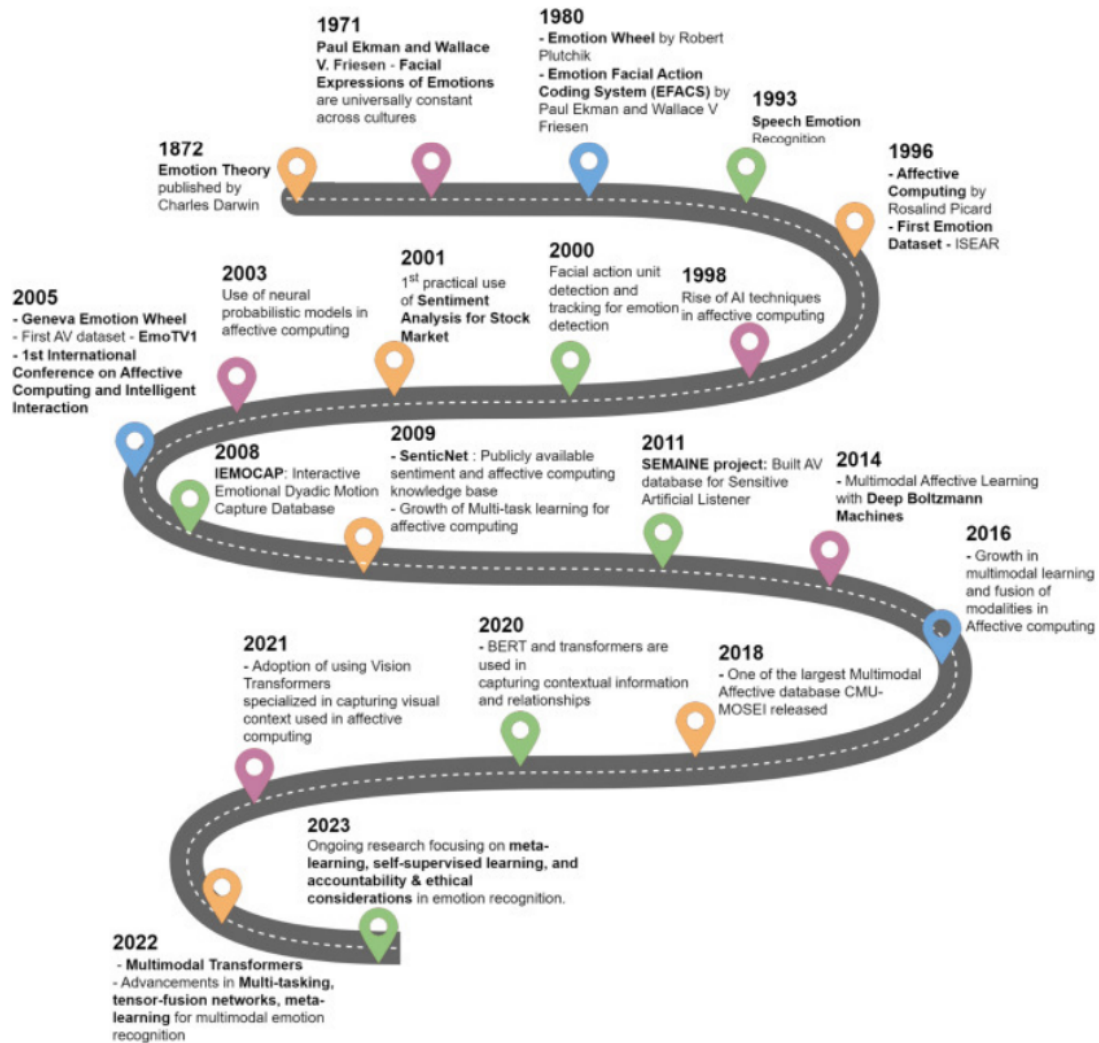
Agent problem is the main problem of artificial intelligence problem, simple agent is something that perceive, believe and act.

2.1 Classic Machine Learning

You have an agent and a human, what you're doing is gating some data such speech data, visual data, physiological data and others. Then we need to extract feature and then do the classification or regression or clustering. What changes now with deep learning is that we don't need to extract the features.



This is a review of use of transform in effective computing, there is a clear selection of topic at least in emotion theory and that at a certain point they started using deep learning:



From this root there are not concerned stuff about emotion theory, this is a sort of biased view.

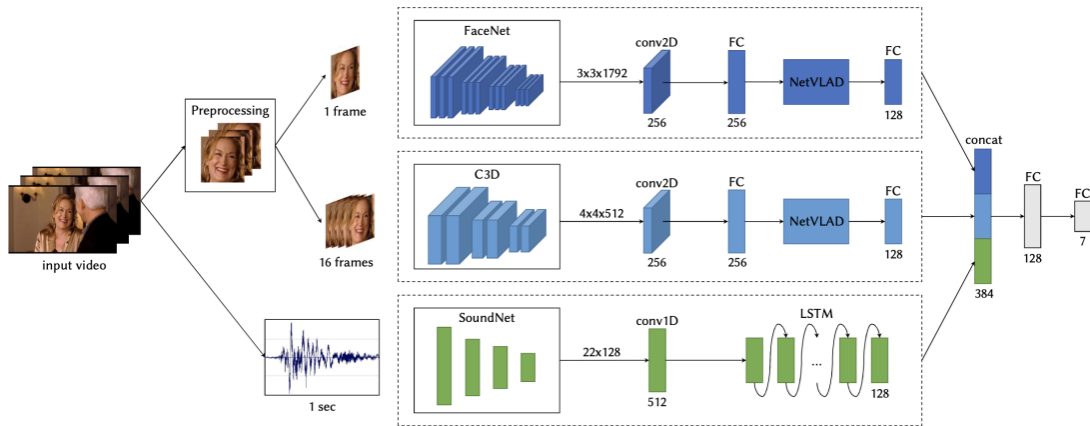
Also an other trend, since the beginning the field was characterize by adding modality, nowadays we are working in a multi-modality problem:



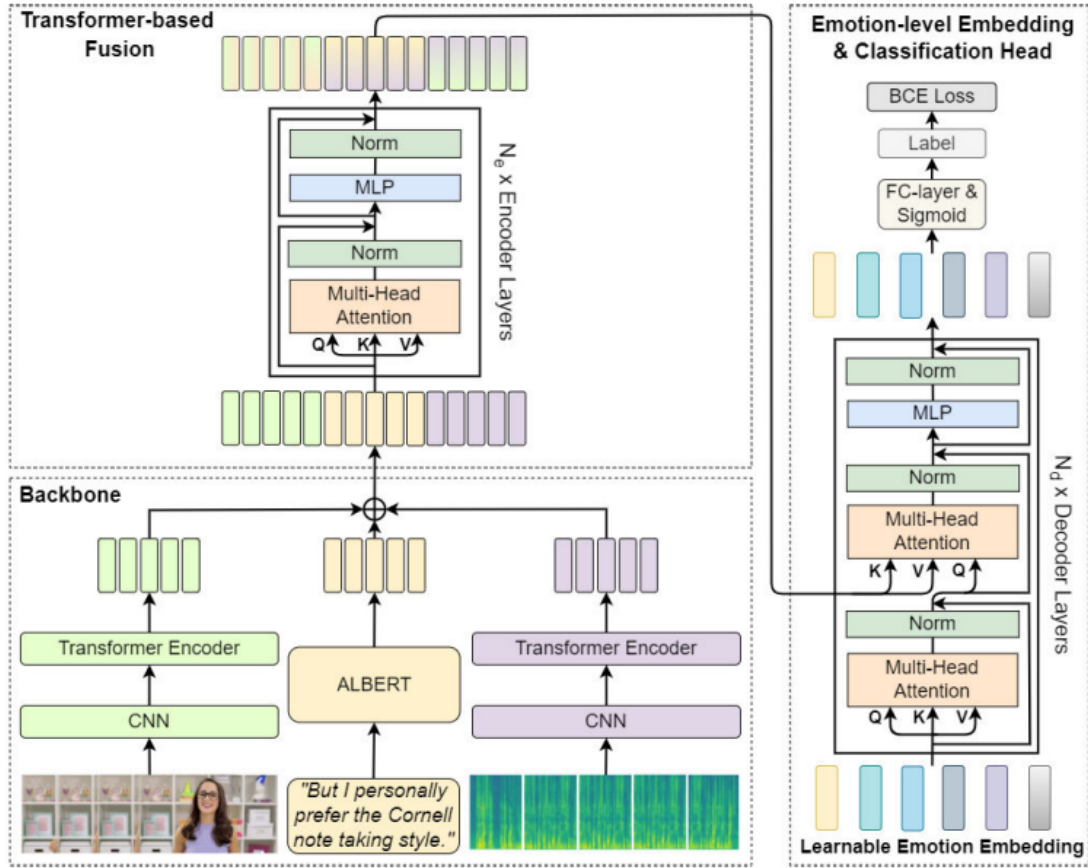
In this effort when you start bring in things like LLM, at least they have some sort of conceptualization of the world, so when you bring them in and mix with the other data, you have a sort of conceptual background which is not something that we had before.

Most of the people put effort in the first problem, detection, also still this bias from the use of facial expression still hold in a certain sense, there is a huge are considering image data from facial expression. Nowadays are increasing the studying of physiology and a multi-modality approach.

Many world we've being dealing in these years we have this end to end architecture:



The things may be complicated when we introduce some temporal models, like attention, fusing different modality:



Here we are injecting some concepts using language model, this change a little bit the game.

2.2 Classical AI Way

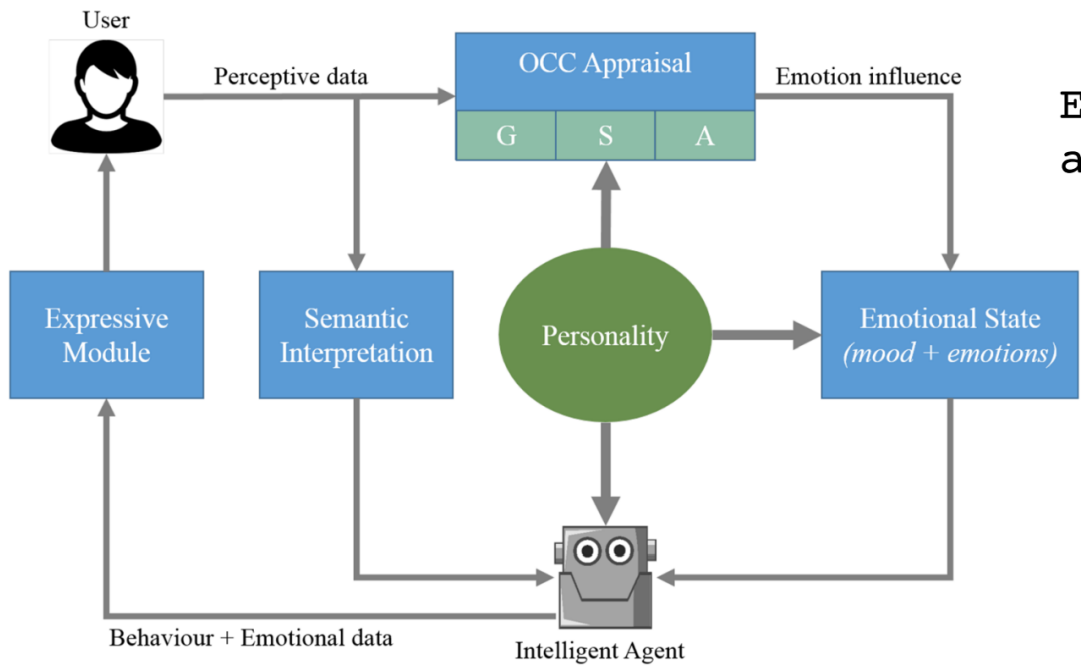
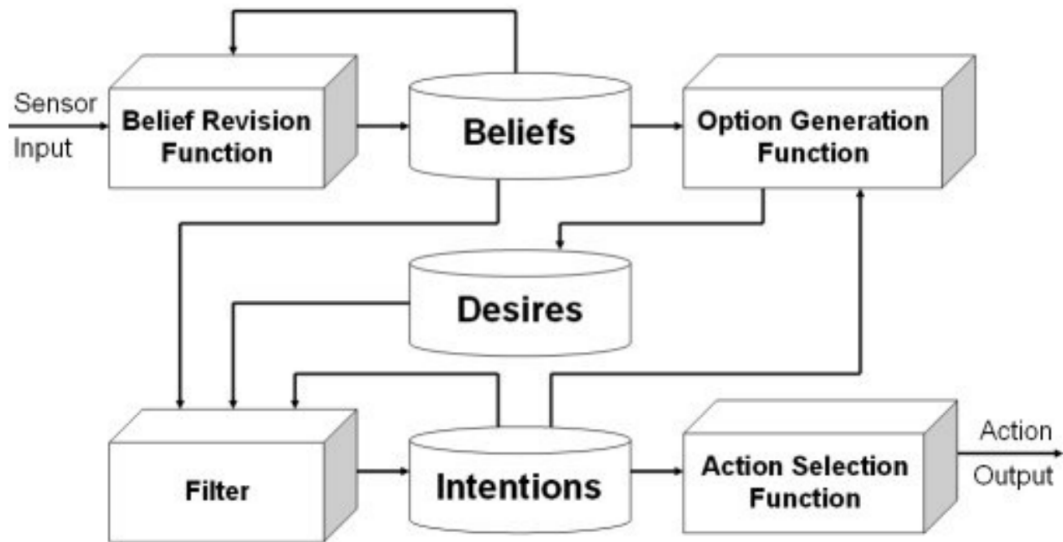
This architecture help us to understand the BDI model (believe, desire, intention).

- Believes: I must have a logical proposition where the door is an object a boolean value that holds the state of the object. Might be probability about fact or object in the world
- Desires: in the modern term might be the goal, the door is open and we have noise so mi DESIRE would be to close the door. I have to deal with attainable goal
- Intentions: is correspond to plans, sequence of action that I can perform in order to satisfy my goal

We can have some modules that inject some kind of emotion in this model.

We can have a lot of kind of this agent, in the example we have a module that respect the OCC Appraisal that allow the module to describe emotion, then the emotion state is the mood plus some emotion:

We can see a more complex example, we have an external interface and an internal interface, coping is used to mitigate or enunche some emotional state is kind of action



that we put in action to solving some emotional issue. The central part is the appraisal module:

Then we have the WASABI architecture, that is much more complete, at the base we have some kind of pleasure Arousal (core affect), this is the base for setting our emotion dynamics, this return to have primary end secondary emotion and this are going to influence our kind of decision and finally we deliberate an action in the world. In the same time we have to appraise what is happening in the world:

The problem with this architecture is that they are quite old, would be difficult reuse one of these architecture with an other and in these day they have no sense. Some of these have been use in robotics, some in agents for instance in videogames. Robotics provide a really reach landscape, there are a lot of architectures.

This is a system that can pay attention on something using cameras, than we have high level perception in which the agent could live in a world of toy or people, they really implemented the omeostatic equilibrium living in the robot (they feel hungry, tired...).

In the kismet architecture we have the basic core affect that act as releaser in order to trigger some categorical emotion. This emotional part going to manipulate the motor expression. In the behaviour system we have an hierarchy of behaviour for example is more important the social behavior than the toy behavior. Obviously in the base we have the things for survive, in case of robots are battery.

We've seen the case study of the mirroring model, mirroring in sense when you smile while others smile. The idea was to have external action, like expression action, and than we have internal actions that also involves action that involves the visceral (visceromotor simulation), this mimic capability must relay only on the external action but yet we'll still use also the internal part, evolution (or whatever) paired the external action with the internal. The modely they used at the more general level was Adolph's model, central affect stuff that evolve for finding higher level emotion, for instance in facial processing what Adolph says is that in the beginning we have a first triggering of sub-cortical model and than we have a low level route which is going directly to the amygdala we have a quick evaluation of the importance of stimulus. Then the amygdala triggers the hypothalamus the autonomic facial control system. The main goal is not to address the high level emotion but to achieve the affective part that regulate the emotion, there are three region that are necessary:

- Amigdala
- Orbital frontal cortex
- Insula

Now they had a problem of simulation, they had to go back to Damasio's theory, set up an internal simulation.

Let's see the emotional braind, first we have the visual part with sensors, than we have all the central affective control that control emotion, in between we have the visuomotor ruote that act when someone perform an action, represent some kind of action recognition. There is kind a hierarchy where all the levels works in feed forward but also in backward, modulating of prediction.

We can represent this model using a PGM, note that what happens in $t + 1$ depends on what happens in t , and the fact that this system evolve in times means that when I move something at time t everything changes.

Chapter 3

Folk Theories

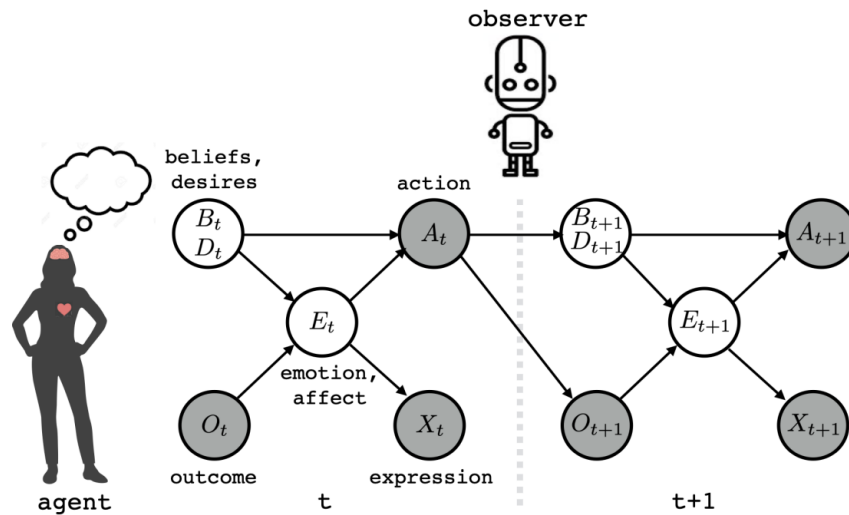
We'll see a different approach, proposed by Goodman (which is a psychologist), the idea is to use folk theories, which have been popular because in everyday life we use folk theories about, for instance, if the speed is high the time when I'll impact the car is less, we do this kind of evaluation everyday, in order to do that we don't have to solve the equation of motion but we have interpretation of motion.

If we have intuitive physics that we use everyday we also have intuitive psychologist. We can interpret the emotion of other people based on facial expression and so one, we don't need to do this interpretation knowing how the others mind are made. It's a kind of instrumental theory that is usefully in the everyday life.

This kind of theory are much more similar of theory of mind approach, agent think that I think.

The idea of this theory is to answer very simple question, what is in the agent that we are observed? what are the believes? what are the events in the world that might cause this believe appear?

This is kind of model that we are proposing:



It's based on Graphical Model, this is a very simple graph. Here we might distinguish two parts:

- Slice at time $t+1$, in the right (to observer)
- Slice at time t

We might collapse nodes in one single node, what we have is that what happens on $t+1$ only depends on what happened on time t , this is called the Markov Model $P(x_{t+1}|x_t)$. This is a very simple assumption, for instance in mind this cannot be true, my believe at time $t+1$ depends on all my previous belief, not only the one at time t .

We can assume in the naif way of this model, that an emotion that has the agent is based on it's beliefs and the outcome of the world. And we know that if the agent is in some certain emotion it'll have an expression and perform certain action.

Note that based on PGM we see the gray node that are observed and the white that are not.

At time $t+1$ the action at time t is the new outcome that we observe and also create the new beliefs.

What does it stand this graphical model? we can see in two way:

- represent the theory of the observer has of the agent. All this stuff is in the head of the observer in order to interpret what is in the head of the agent.
- the process theory of the agent

This simple model allows to perform a number of inferences, they setup taxonomies of inferences. This models are generative models. In statistics the best generative model is SAMPLING, we have a population and we choose some individual of this population, but how do we choose this sample? behind the sampling we have a distribution, we sampling some x from the distribution. One other very simple distribution is the Bernoulli (the toss of a coin), when we sample on this distribution we can obtain 1 or 0.

The tilde sign in statistic means that the random variable x is distributed as the distribution, but when we use it in the case of sampling we mean that the variable could be 0 or 1. How the generative model works:

- I sample a variable x using the distribution of probability of x : $x^* \sim P(x)$
- than I sample an y from a probability distribution of y given x , and x is the variable that i sampled before: $y \sim P(y|x = 0)$. This conditional probability means that according to the value of x , in this case happy, we produce a y , that is a configuration of action unit (facial expression). This give the probability of activation of action unit given that affective state is happy.

So in this way we have a generative model, we sample, this goes under the name of Montecarlo Sampling. We did this stuff because we've seen a generative model, for instance belief and desire depends on action, so we have a conditional probability that generate the new action given believe and desire:

$$B_t, D_t \sim P(B_t, D_t|A_{t-1})$$

Then next sampling is the outcome, the future outcome depends on the action at time $t-1$, again we have an other conditional probability:

$$O_t \sim P(O_t|A_{t-1})$$

And again we can sample the emotion, given B , D , O :

$$E_t \sim P(E_t|B_t, D_t, O_t)$$

Than we can sample the expression again:

$$X_t \sim P(X_t|E_t)$$

And finally the action again:

$$A_t \sim P(A_t|B_t, D_t, E_t)$$

This is the essence of generative model, when we specify the graphical model we specify the type of the problem.

It's interesting to understand what we can infer.

3.0.1 Recognition

Model can be used for recognition problem, for instance given an expression it can understand the probability that this emotion is given by a certain expression $P(E_t|X_t)$. Or given an action we want to know which emotion will be generated: $P(E_t|A_t)$.

3.1 3-rd Person Appraisal

It can be used also for a kind of 3 person appraisal. What kind of understanding is trying to perform? Let's look the conditional probability, I agent I've seen a certain outcome and I'm trying to infer what is the state of the user (I'm trying to appraise the emotional state of third person):

$$P(E_t|O_t)$$

Then I'm asking what is the affective state giving the certain outcome but given also the most likely belief that the person has:

$$P(E_t|O_t, B_t, D_t)$$

In the last we is more complex because we don't observe outcome and we have to infer from beliefs and desires what a third person is feeling:

$$P(E_t|B_t, D_t)$$

3.2 Causes

Then we can infer the causes, back on reason. I know that it's in some state, now I have to conjecture why it's in this state, I'm trying to infer the cause of the current state:

$$P(O_t|E_t)$$

$$P(O_t|E_t, B_t, D_t)$$

We can complicate this inferring possible outcome given Emotion and Beliefs: It's a kind of backward reasoning because we are going back:

$$P(O_t|E_t) = \frac{P(E_t|O_t)P(O_t)}{P(E_t)}$$

3.3 Cue integration

Then typically engineering, but also efficiency problem, I might integrate cue:

$$P(E_t|O_t, X_t)$$

$$P(E_t|O_t, A_t)$$

$$P(E_t|X_t, A_t)$$

$$P(E_t|O_t, B_t, D_t, X_t, A_t)$$

The problem is how to put all the factors together, we can do that empirically, but we might have some principal way to design this probability conditioned by multiple cue, the tool is perform multi-modal integration.

3.4 Reverse appraisal

Then we can use the model for reverse appraisal, given a certain effective state what was the belief and desire that take it to that effective state:

$$\frac{P(B_t, D_t|E_t)}{P(B_t, D_t|O_t, E_t)}$$

3.5 Prediction

Then prediction, if it's in certain effective state which kind of action it'll do:

$$\frac{P(A_t|E_t)}{P(X_t|E_t)}$$

The posterior predictive is to confront my prediction, used monte-carlo, with actual data, if there is some error I'll use the error to adjust the parameter of distribution to minimize the error:

$$A_t \sim P(A_t|E_t = e)$$

3.6 Counter-factual reasoning (true causal)

We can use the model for the counter-factual reasoning, the idea is what kind of emotional would have been if an action hasn't occurred, this is the base for the causal reasoning:

$$\frac{P(E_t|\neg O_t)}{P(X_t|\neg E_t)}$$

3.7 PPL

The idea of probabilistic programming language is to build on some kind of framework (for instance pytorch) the stuff that we said before.

Are language that allow to produce program as we did before. There are two primitives:

- Sample
- Observe

In the pyro example we see that we read some data than in the model we sample a parameter and we use this parameter in the distribution.

```
def model(data, params):
    # initialize data
    N = data["N"]
    y = data["y"]
    with pyro.plate("data", N):
        theta = pyro.sample("theta", dist.Beta(1., 1.))
        pyro.sample('obs', dist.Bernoulli(theta), obs=y)
```



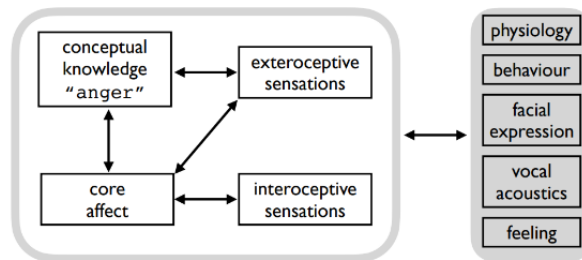
$$\theta \sim P(\theta)$$

$$x \sim P(x \mid \theta)$$

Chapter 4

Perception

Perception is something trickier than what it seems. Emotion according to conceptual theory is a revolution of appraisal theory, but here we have a really deep understanding.

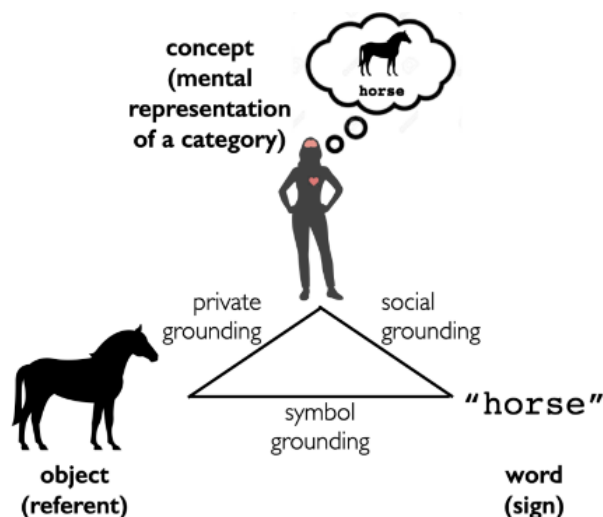


This is the basic scheme:

- We have something in our mind because we've learned before, we have some concepts
- External perception
- Interceptive: Internal perception: perception of our body
- Core affect: is the same as Russell core affect, all other stuff concur to put the body in a certain state

Emotion is a conceptual, anger is a concept. It's not a one way stream but exactly the opposite I recognize something as an emotion because I'm trying to predict that emotion. Our brain always predict.

4.1 Categorical Perceptions

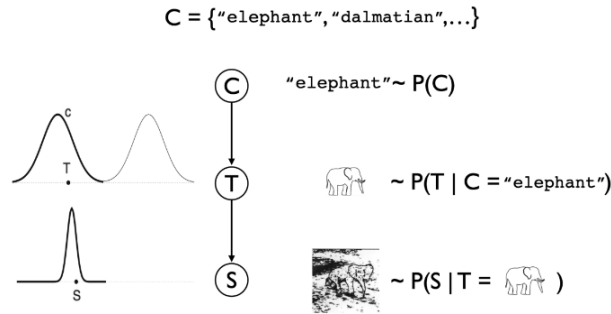


At the very basic level there is theory of concept. Since we using language we might tight with is name. Concept is a mental representation of a category (horse). Given a concept our brain give us a representation, concept is the mental representation of the semantic of that name.

Social grounding is allowed by our language, we agree on concept because it's a common sense.

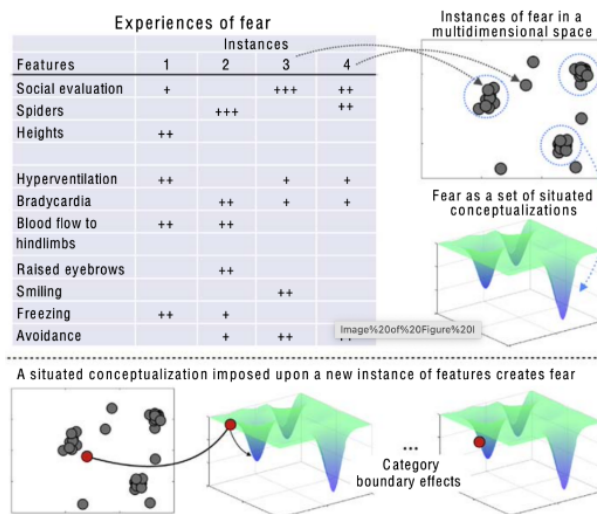
For instance we have colours, colours are a discretization of a continuous concept. We continuous categorizing things.

The voice perception is called the magnet effect, the facial expression are continuous but we know that we have group of neutral expression and a group of anger expression and some uncertainly. We have some kind of binary categorization even if the facial expressions are continuous.



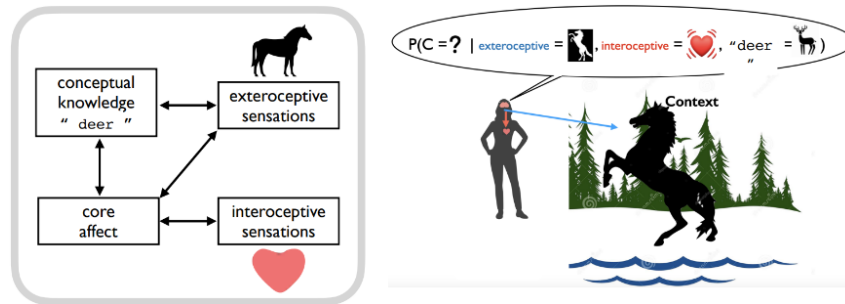
The idea in simple terms, in PGM terms, we have a random variable representing the category (a list), I've a probability of each category, than we have to sampling over the distribution of probability, but what we sample? here we sampled the target, it's the idea of category (for instance the idea of elephant). Then given that I sampled that shape we sampled the image that we see using the conditional probability in order to generate the image. Then to use this to infer we need to use Montecarlo sampling in order to change our model of the world. We can substitute this simple stimulus with internal and external perception in order to understand how we understand emotions.

4.1.1 Emotion is a category



Consider for instance one category, limit to fear. We take some numbers to evaluate some items that describe that category and than we can do some different experiment

sign a + when each feature occurs. We then see that some features occur more than other in different context. We can put some feature in a cluster and this represent a kind of fear cluster, different fears can be represent by different clusters. This tell us that what we say as fear is a ensemble of concepts. That's why from practical point of view, even if we know the label, recognize emotions in experiments. We can achieve a good prediction for a person but it's really difficult to generalize.

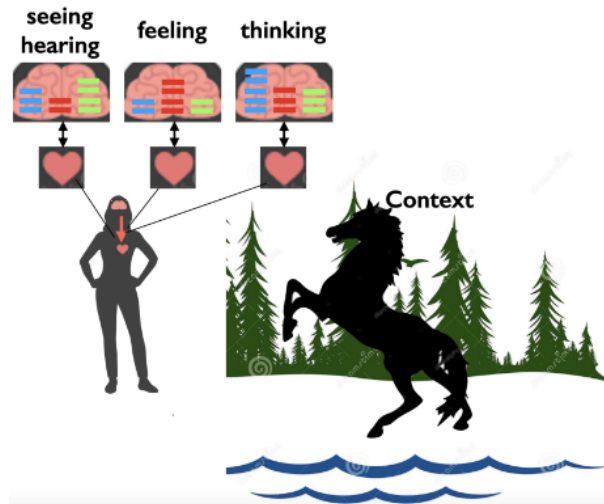


Any categories can be mentally represented according to this scheme (the one we saw before), but actually we have mental memory of the category that we're dealing, for instance if we only have a deer category and we see an horse for the first time we can only see a deer. In this case at the beginning we have also some kind of internal affect representation. And we are trying to match all this in order to predict the concept.

This mechanism of conceptualisation is a mechanism that we always use. When we use in psychology terms like emotion, cognition, perception are just category to make things clear, but there is no real distinction (there is nothing in our brain that is emotion or cognition). When we speak we have to define concept, when we use world this world become a real thing in our brain, this world has a meaning but we cannot assume that this world has a specific part in our brain.

4.1.2 Mental states are categories

What we have in our brain is a complex network of networks. We have some fundamental large network like the attention, the salience, these are very large and part of the area of a network might serve an other network. We can take a photo of brain showing which part, and which networks, of network is active when we feel (or perceive) something. What we can see is that every photo are quite similar to others, so we cannot say that in one moment we are thinking and so on.

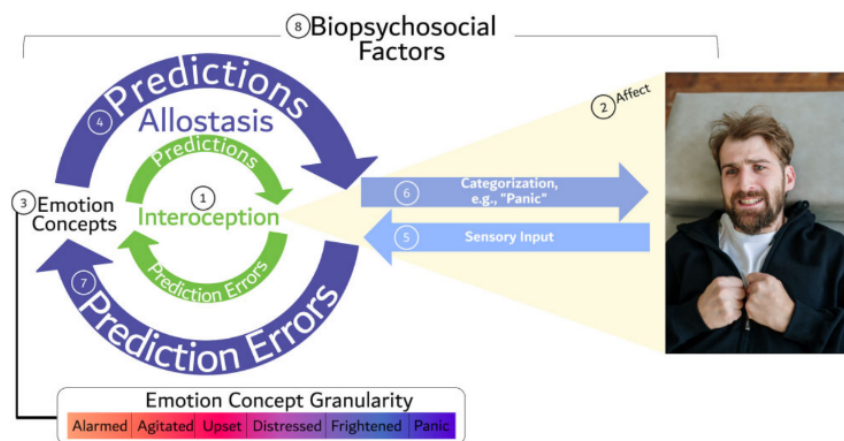


4.1.3 Categories are in service of allostasis

In the base of all of this there is the brain and the body. The story in the end is very simple, we are trying to keep our self alive, we are trying to regulate our internal environment to be alive. There is a kind of homeostasis in our body, we have some threshold that regulate our internal state, we are predicting the homeostatic state, this is called allostasis.

If we have more knowledge we can do this more efficient, for instance if we have a very clear understand of emotion (granularity) than we are more effective in predicting our state and regulating our state (we have more word to describe our state).

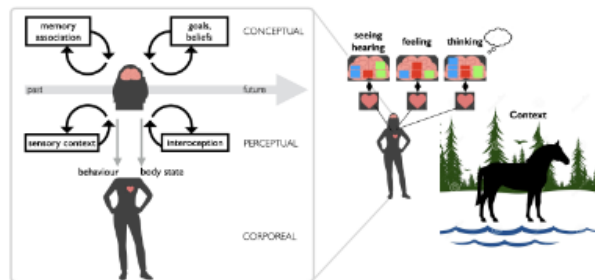
What happens to people that have a low emotion concept granularity we cannot feel a lot of emotions. There is really correlation between our language ability and the ability to recognize our effective state, this because we live in a words of language category.



Chapter 5

Predictive Brain

It's important to see that in the brain we don't have the thinking part, but what we have is a number of basic networks that are differently weighted.



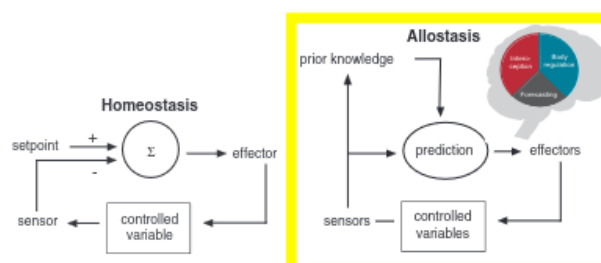
We have a lot of hypothesis:

1. The brain is not for thinking
2. There is one and only brain
3. The brain develops by wiring itself to its world
4. The agent's brain predicts what the agent does
5. A brain works with other brains
6. One brain makes more than one kind of mind
7. Brains can create reality

5.1 Allostasis

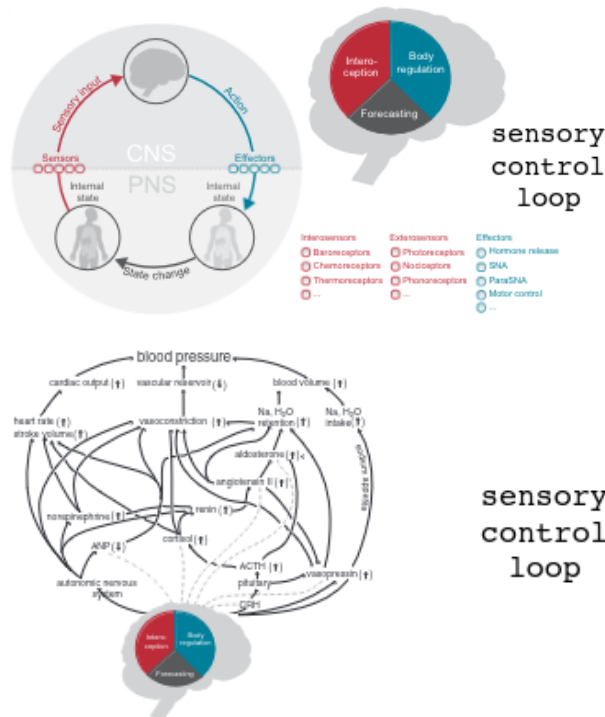
In the basis all the networks have not other purpose than keep the individual alive. We have some threshold under which the body is working correctly, how do we regulate our body? Homeostasis control our state, we have a set point and when we are over or below that level we have regulation. In order to have this regulation we put the derivate of the output as new input.

We can also predict our state, we have some prior knowledge and we can use this to drive our state. This mechanism is called allostasis, that is the base of what the brain does.



We can see a classical control loop:

But actually what we know is that we have prediction, in the example we have the simplified map of blood pressure control. What we can see is that is a very complex system in which we have some mechanical controls:



5.2 There is only one brain

There are some psychologist that think that we have multiple brain:

- Reptilian
- Limbic brain: locus of the emotions
- Neptocortex

This is completely false, there is not scientific evidence.

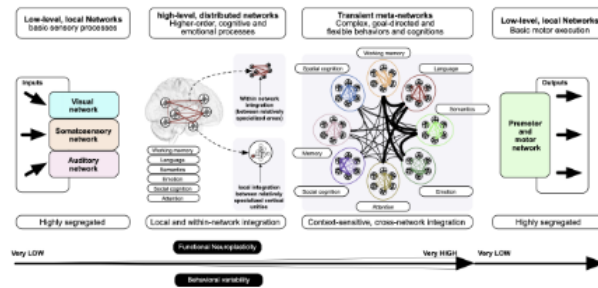
5.3 The brain is a network

What we have actually is that the brain is very large and complex network which is made up by large sub-network. This interleaving of networks is always actual. We have some basic interfaces network:

- Visual network
- Somatosensory
- Auditory

Basically in our brain we have regions that are connected. When we go to higher order distributed network we are dealing with higher network.

Actually our brain is always active even when we are doing nothing. Doing resting experiments they discovered the default network that is always active. It's always active



because to answer quickly to all the possible stimulus. From this we can get the complexity of brain, there is not a specific region but there are overlaps.

The network collaborates:

Chapter 6

Approximate Inference

We can have different models with hidden variables:

- hidden states: z
- parameters: θ_z, θ_D
- model itself

We want compute:

$$P(h|D) = \frac{P(D|h)P(h)}{P(D)}$$

$P(h)$ is a prior distribution.

Obviously we might have the problem to infer among all the hidden variables, in this case we are performing what in machine learning is called learning (infer the parameters). In machine learning learning is to infer the parameters: $P(\theta|...)$ and inference is $P(z|...)$. From the Bayesian point of view we are always inferring the variable.

In the worst case everything we observe are the data, we are learning from the data.

h are just hypothesis on the world, given this we assign some probability on what we observe in the world, according to that we revise our probability the world. This has been use to formalize the process of scientific review.

$P(D)$ is the bottleneck of bayesian inference, this could be obtained through marginalization:

$$P(D) = \int_h P(D, h) dh$$

The problem is that is very high dimensionality with many variables. Another way to define $P(D)$ is the partition function, is a definition that comes from statistical mechanic, we have a gas and a space, this is the function that we might calculate summing over all the possible states of the particles of the gas. It's incomputable, so we have to approximate. From that derive a lot of stuff:

$$\int_h P(D|h)P(h)dh$$

For instance we want to compute the expectation, the expected value is the mean:

$$E[F(f)|h] = \int_h F(h)P(h|D)dh$$

In this case we want to know the expectation of a function (function of heights of people) we have to integrate that function with respect to the normal distribution (gaussian). But for the inference problem the distribution that we have to take is the posterior, that we have to compute using the bayes rule and use the partition function, that again is not computable.

We have another problem, the marginals over hidden few variable, we want to compute the probability of a set of variables, that mean that we have to marginalize:

$$P(h_i|D) = \int P(h_i, h_{-i}) dh_{-1}$$

Final problem is the prediction, compute the probability of a new theta given the theta that we are observing up to now:

$$P(d^{new}|D) = \int P(d^{new}|h)P(h|D)dh$$

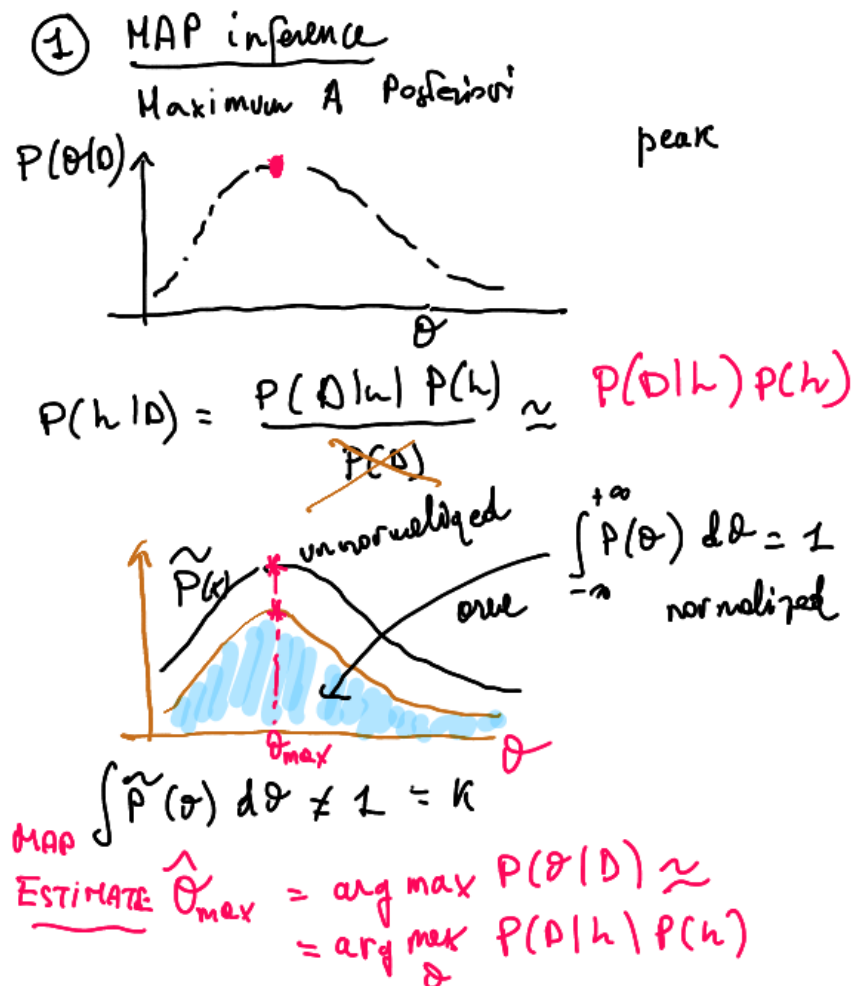
6.1 Approximation

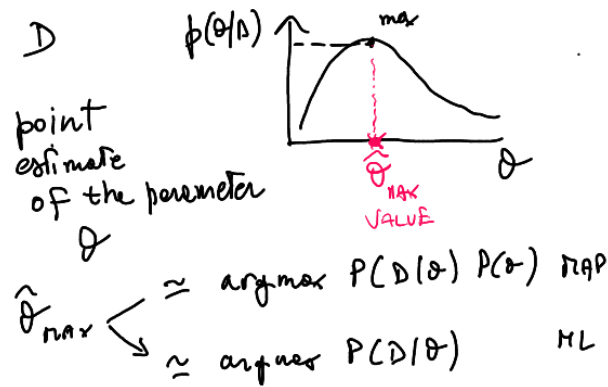
Given those problems we have a stack of approximations.

The very simple approximation is called the Map Inference (Maximum A Posteriori). Assume that we have a distribution over a parameter, since we are interested in the maximum what we can do is that we can approximate in which we discard the normalization factor $P(D)$. So we now have a non normal distribution (doesn't sum to 1) but we can use the maximum as an approximation of the maximum normalized.

Map Estimate:

$$\begin{aligned}\theta_{MAX} &= \arg \text{MAX} P(\theta|D) \sim \\ &= \arg \max_{\theta} P(D|\theta)P(\theta)\end{aligned}$$





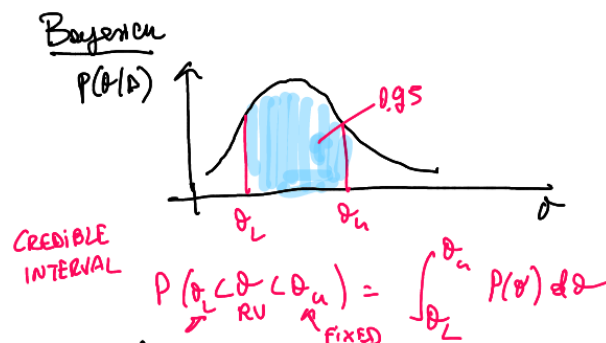
Understanding means that we have to estimate the error:

$$P(\theta_L \leq \theta \leq \theta_U) = 0.95$$

Let's do an example, the heights. For instance we have God that extract out height using a distribution, the probability of height is theta and it's unknown. What we can estimate is the sample, we can estimate also an interval that stands between a lower bound and an upper bound in which most probably the theoretical theta falls in.

The important thing is how do we write this parameter, we want to compute the probability that our theoretical theta is in the interval. This means that from the Bayesian point we know that distribution of our theta given the data and we are searching that could be an interval that covers most of probability:

$$P(\theta_L \leq \theta \leq \theta_U) = 0.95 = \int_{\theta_L}^{\theta_U} P(\theta) d\theta$$



In classical statistic we can say:

$$P(\theta_L \leq \theta \leq \theta_U) = 1 - \alpha$$

But they are completely different, in the first case we have the probability that theta, a random variable, are between two limits that are fixed. In the second case the random variable is the interval, and theta is fixed (unknown but fix), so the best interpretation of the classical statement is that you have a theta space and you have a value of theta that is a number and we do experiments and in all experiment we determine the theta upper and lower (the confidence interval), some of these confidence interval will contain theta:

6.2 Laplace approximation (saddle point approximation)

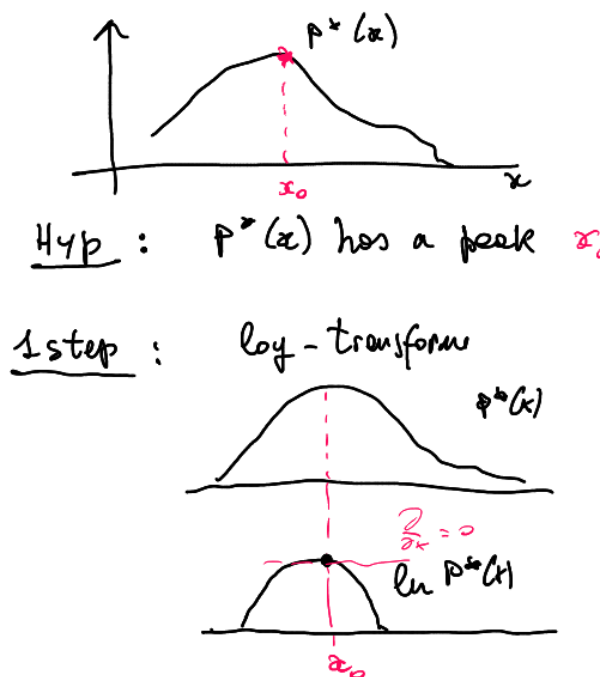
It's very useful and effective. Let's do first in 1D case. Assume that we have some sort of unnormalized distribution, means that if we take the integral of this distribution this integral is different from 1, the integral is the partition function, which in fact provide the way of normalize:

$$\int_{-\infty}^{\infty} P^*(x) dx = Z_p$$

$$P(x) = \frac{P^*(x)}{Z_p}$$

The basic hypothesis is that p^* of x has a peak (kind of a maximum), and this peak might be in x_0 .

The first step of this approximation is taking the log transform because the log is a monotonic increasing function:



The second step is the Taylor series expansion around x_0 :

$$\begin{aligned}
\ln P^*(x) &= \ln P^*(x_0) + (x - x_0) \frac{\partial}{\partial x} \ln P^*(x)|_{x=x_0} \\
&\quad + \frac{1}{2}(x - x_0)^2 \frac{\partial^2}{\partial x^2} \ln P^*(x)|_{x=x_0} \\
&= \ln P^*(x_0) + \frac{1}{2}(x - x_0)^2 \frac{\partial^2}{\partial x^2} \ln P^*(x)|_{x=x_0} \\
\ln P^*(x) &= \ln P^*(x_0) - \frac{1}{2}C(x - x_0)^2 \\
e^{\ln P^*(x)} &= e^{\ln P^*(x_0)} e^{-\frac{1}{2}C(x-x_0)^2} \\
P^*(x) &= P^*(x_0) e^{-\frac{1}{2}C(x-x_0)^2}
\end{aligned}$$

$$\begin{aligned}
Z_p &= \int p^*(x) dx \approx \int Q(x) dx \\
&= \int p^*(x_0) e^{-\frac{c}{2}(x-x_0)^2} dx \\
&= p^*(x_0) \int_{-\infty}^{+\infty} e^{-\frac{c}{2}(x-x_0)^2} dx \\
&= p^*(x_0) \sqrt{\frac{2\pi}{c}} \\
c = \text{precision} &= \frac{1}{\sigma^2} = -\frac{\partial^2}{\partial x^2} \ln p^*(x).
\end{aligned}$$

We know that C is the inverse of the variance square, the precision is kind a geometry quantity. This fact seems innocent but give a clear correlation between uncertainty and geometry. We don't know the original distribution but we now know that it's kind of a Gaussian.

Now we can work in term of credible interval because we have a Gaussian distribution. Now we have a distribution and this is very important because there are many theory based on this possible solution.

Also this opens up a number of possibilities, by the way this can easy be extended to D grater than 1 dimension, basically we have the same approximation but using a vector (x_0 is a vector) and the Hessian matrix that is basically the mix:

$$\begin{aligned}
\ln P^*(\underline{x}) &\sim \ln P^*(x_0) + \frac{1}{2}(\underline{x} - \underline{x_0})H(\underline{x} - \underline{x_0}) + (\underline{x} - x_0) \\
H &= \frac{\partial^2}{\partial x_i \partial x_j}
\end{aligned}$$

6.3 Monte Carlo

Again we have an x that can contain hidden variable, parameters and other things, anything we want to put in distributional. Than we have $P(x)$ that represent the distribution

of x . We know the integral of $P(x)$ is normalized that means that is equal to 1. And we know $\tilde{P}(x)$ of x that is unnormalized this means that :

$$P(x) = \frac{\tilde{P}(x)}{Z_p}$$

$$\frac{\tilde{P}(x)}{\int \tilde{P}(x) dx}$$

Under this settings we have two problems, the first is sampling, we want to sample some points from the distribution:

$$x^{(r)} \tilde{P}(x) r = 1, \dots, R$$

The second problem is to compute expectations, for instance mean, this is less problematic if I know how to sample. In order to compute variance we have to compute expectation and to compute expectation we need to compute integral

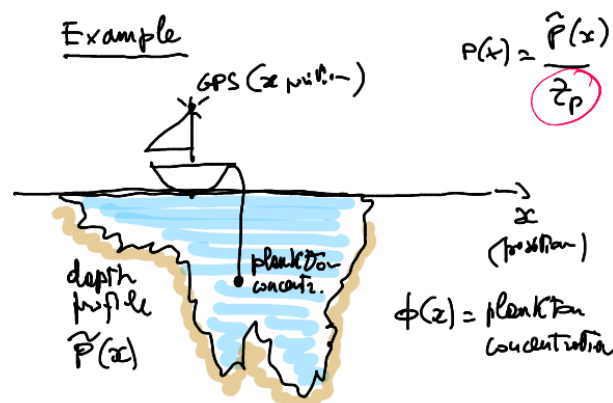
$$\Phi(x)$$

$$\langle \phi(x) \rangle = \mathbb{E}[\Phi(x)]$$

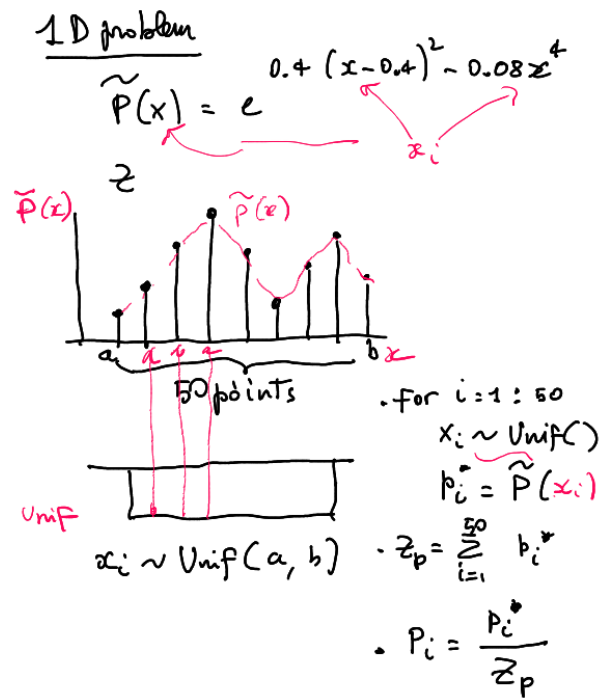
$$= \int \phi(x) P(x) dx$$

$$\text{var}(\phi(x)) = \mathbb{E}[\phi(x)^2] - (\mathbb{E}[\phi(x)])^2$$

The very hard problem is sampling, let's do a physical example. Assume we are on a lake which has a certain depth, we are on a boat that has GPS and we know approximately the depth profile of the lake that is the unnormalized function $\tilde{P}(x)$ and we have an instrument that measure the plankton concentration that is the function $\phi(x)$. In a sense our research job is sampling a position that is the problem 1 and we have to compute the average plankton concentration which is our expectation $\langle \phi \rangle = \frac{1}{Z} \int \phi(x) \tilde{P}(x) dx$. Obviously we cannot do a complete sampling but we have to do a choose of position, we can try a sort of uniform sampling.



Assume that we know the tilde, might be a function, but we need to know the partition function because is used to normalize. We can choose the sampling points using a uniform distribution:



If we move in a 2D space we have the double of states, because we have to sample the x and y (latitude and longitude of the boat). Clearly when we are dealing with ML the number of variables are really high, for instance are the number of the features. So when we go up with this condition is not unusual have 1000 dimensions that means that in this case we have 50^{1000} states (50 because are the sampling points).

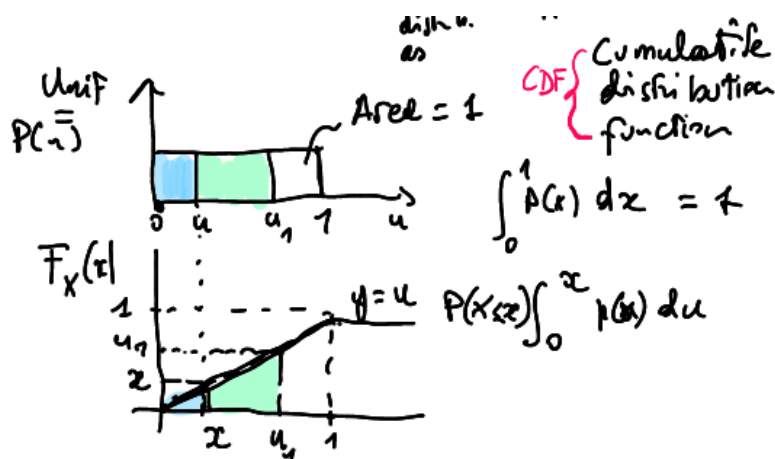
If you consider the universe are estimate 2^{266} electrons and 1 electrons may be equivalent to a 1000 GHz processor that can compute 2^{40} states per seconds. Now consider as a constant the age of the Universe that is 2^{58} seconds. Now let say that for all this time and for all this processors we will be able to compute 2^{354} states that is way less then the 50^{1000} . So it's impossible to compute all this sample. This is way we need some clever way of sampling.

A brief recap, we are trying to have a way to sampling. For instance we can sample from a uniform distribution, assume that we have a random function that return a random number between 0 and 1. Given that we might have different techniques.

6.3.1 Inverse Sampling

The first one we might consider is inverse sampling. Assume we have a random variable X and we can describe this random variable with its distribution function that describe all the value up to the value X , this is called cumulative distribution function (this mean that we are going from 0 to 1 and accumulate, growing function, the sum must be 1 at the end by definition):

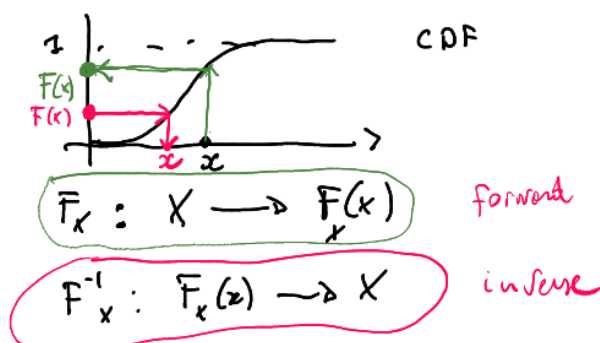
$$X \sim F_x(x) = P(X \leq x)$$



This means that the cumulative function of uniform distribution is just u , because it's a line $F_{unif}(u) = P(X \leq u) = u$. Now assume that we have a random variable and this cumulative function CDF:

$$X \text{ R.V. } F_x(x) = P(X \leq x) \text{ CDF}$$

We now need an hypothesis, must exist the inverse of the CDF $\exists F_x^{-1}(x)$.



The inverse function means that if I choose a value of F we go back and obtain the x value. We can use this inverse sampling with distributions that have this inverse function. The idea of inverse sampling is that we can sample x from $P(x)$ and that is equivalent to execute this problem: sampling a uniform number from a uniform distribution then just plug this uniform number within the inverse distribution function until the condition:

$$x \sim P(x) \leftrightarrow \text{repeat } \left\{ \begin{array}{l} u \sim \text{Unif}(0, 1) \\ x = F_x^{-1}(u) \end{array} \right\}$$

Example:

$$\text{Exp}(x \mid \lambda) = \lambda e^{-\lambda x}, \quad x > 0$$

$$F_{\text{Exp}}(x) = 1 - e^{-\lambda x}$$

Find inverse CDF:

$$y = 1 - e^{-\lambda x}$$

$$\implies e^{-\lambda x} = 1 - y$$

$$\implies \ln(e^{-\lambda x}) = \ln(1 - y)$$

$$\implies -\lambda x = \ln(1 - y)$$

$$\implies x = -\frac{\ln(1 - y)}{\lambda}$$

$$F^{-1}(u) = -\frac{\ln(1 - u)}{\lambda}$$

$$u^* \sim \text{Unif}(0, 1)$$

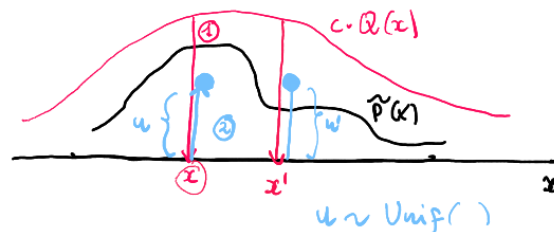
$$x = -\frac{\ln(1 - u^*)}{\lambda}$$

In the Exp example we've seen that we have a distribution of what we know the form, and we want to sample from this distribution. We used the inverse sampling in order to approximate the distribution. The idea is that if I can find the inverse of the function we can proceed with the inverse sampling, we can do this sampling n time and obtain n approximation. If we do this sampling a lot of time we can find an overlap of the original distribution, we can prove it but we'll not do.

6.3.2 Rejection sampling

The next step is the rejection sampling, for instance we don't know the inverse function $\nexists F_x^{-1}$.

The idea is to define a proposal distribution, or choose, and again we have to randomly sampling from a normal distribution. If we choose a Gaussian distribution if we sample a lot of time we'll obtain a gaussian form. So again we have to define a Q from which we can sample, this Q is called the approximating distribution. In rejection sampling we define a constant $C \geq 1 \rightarrow c \cdot Q(x) \geq P(x)$, what we are doing is choosing a Q function multiplying by a constant in order to normalize the unnormalized distribution $c \geq 1 \implies c \cdot Q(x) \geq \tilde{P}(x)$:



First step I sample from Q and obtain an x, the second step is to sample a random hight u from the uniform distribution, now what happens is that I have a point and if the

point is in the are of the P tilde (is under the P tilde) I accept the x point that I sampled before, the other case that might happen is that I sample the second x and sampling the u the obtained point is over the P tilde so we cannot accept the second x . The rule that I'm applying is to multiply the u time c time $Q(x)$ and if it is lower than P tilde I'll accept:

$$\begin{aligned} u \cdot C \cdot Q(c) &\leq P(\tilde{x}) \text{ ACCEPT} \\ u \cdot C \cdot Q(c) &\geq P(\tilde{x}) \text{ REJECT} \end{aligned}$$

The algorithm is the follow, we sample a x from $Q(x)$ than we sample a u from uniform distribution 0, 1 and we do this until we have this condition, the sampled u is greater then P tilde, after all this cycle we've sampled one point:

$$\begin{aligned} &\text{repeat } \{ \\ &\quad x \sim Q(x) \\ &\quad u \sim Unif(0, 1) \\ &\quad \} u > \frac{P(x)}{cQ(x)} \end{aligned}$$

6.3.3 Importance Sampling

An other method is the importance sampling that is a general class of methods that actually do not solve the sampling problem but anyway we manage to approximate computing expectation.

We use importance sampling when we can solve problem 1, sampling from a distribution, and we are interesting in solving problem 2, that is computing expectation.

In general approach in MonteCarlo, we assume that we can sample:

$$\begin{aligned} \eta &= E[\phi(x)] = \\ &= \int \phi(x)P(x)dx \\ &\approx \frac{1}{N} \sum_{n=1}^N \phi(x_n) \quad x_1, \dots, x_n \sim P(x) \\ &= \hat{\eta} \end{aligned}$$

We assume that the N is large and in this case what we have for large sample is the central limit theorem (CLT) holds. CLT implies that the difference between the sampled n and the n is distributed as a Normal distribution, when the variance is the theoretical variance divided by N . Whit this assumption we can find a confidence interval for the sampled n :

$$P(\tilde{\eta} - 1.96 \frac{\tilde{\sigma}}{\sqrt{n}} < n < \tilde{\eta} + 1.96 \frac{\tilde{\sigma}}{\sqrt{n}}) = 0.95$$

We are now using importance sampling avoiding this complex distribution:

$$P(x) = \frac{P(\tilde{x})}{Zp}$$

We introduce the approximating distribution, or the proposal distribution, we know how to sample from this distribution:

$$Q(x) = \frac{Q(x)}{Z_p}$$

In this setting the problem 2 is to compute our expectation of function ϕ , for definition we should:

$$E[\phi(x)] = \int \phi(x) P(x) dx$$

Now we introduce the $Q(x)$:

$$\int \phi(x) P(x) \frac{Q(x)}{Q(x)} dx$$

Now let's make an other trick, divide all this by one, this one could be written as the integral of $P(x)$ because it's normalized by definition:

$$\frac{\int \phi(x) \frac{P(x)}{Q(x)} Q(x) dx}{1 = \int P(x) dx}$$

We can now rewrite all this to introduce the normalized distribution:

$$\begin{aligned} & \frac{\int \phi(x) \frac{P(x)}{Q(x)} Q(x) dx}{\int \frac{P(x)}{Q(x)} Q(x) dx} \\ & \frac{\int \phi(x) \frac{\tilde{P}(x)}{\tilde{Q}(x)} Q(x) dx}{\int \frac{\tilde{P}(x)}{\tilde{Q}(x)} Q(x) dx} \\ & \frac{\int (\phi(x) (\frac{\tilde{P}(x)}{\tilde{Q}(x)}) Q(x) dx}{\int (\frac{\tilde{P}(x)}{\tilde{Q}(x)}) Q(x) dx} = \frac{E_Q[\phi(x) \frac{\tilde{P}(x)}{\tilde{Q}(x)}]}{E_Q[\frac{\tilde{P}(x)}{\tilde{Q}(x)}]} \end{aligned}$$

Now we can sample from Q :

$$\approx \frac{\frac{1}{N} \sum_{i=1}^N \phi(x_i) \frac{\tilde{P}(x_i)}{\tilde{Q}(x_i)}}{\frac{1}{N} \sum_{i=1}^N \frac{\tilde{P}(x_i)}{\tilde{Q}(x_i)}}$$

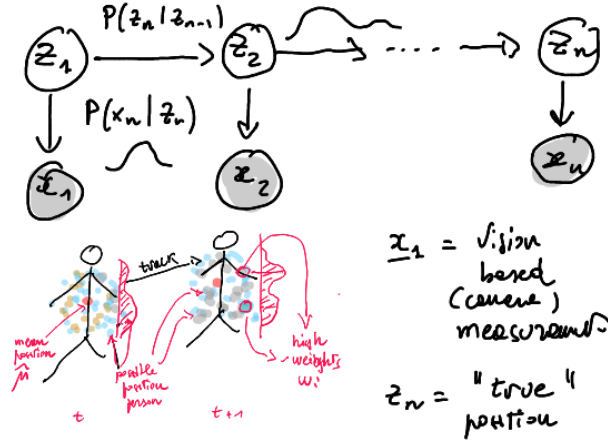
We can call $\frac{\tilde{P}(x)}{\tilde{Q}(x)}$ the importance weights w_i :

$$\frac{\sum_{i=1}^N w_i \phi(x_i)}{\sum_{i=1}^N w_i}$$

We are computing a weighted mean. They are weighting each particle with some mean, this mean that there are particles with less or more importance. Clearly what we've obtained is an approximation of the original problem.

Bayesian Filter

We can see the example of the Bayesian Filter (of which Kalman Filter is an instance). We have a latent state and observations:



The x are observation of positions in space, the z is the true position or not.

The idea of filtering is to compute the true position given all the previous observations, given all the story we want to know the probability to do a certain observation. Obviously what we do here is to simplify, we take Markov assumption and we know that what happens at time t only depends on what happens at time $t-1$.

So again our problem is to compute the expectation over the inferred states, we want to infer the mean position of all particles:

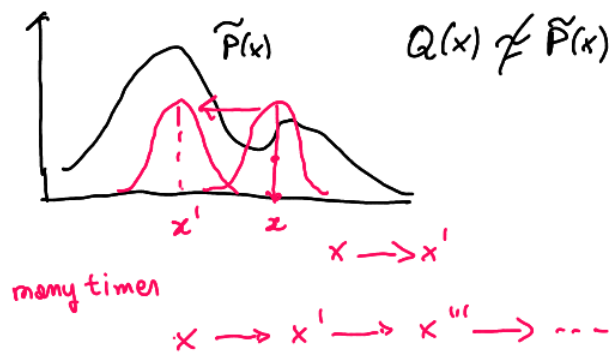
$$\begin{aligned}
 E[\phi(x)] &= \int \phi(x_n) P(z_n | x_{1:n}) dz_n = \text{filtering} \\
 &\approx \frac{\sum_{i=1}^n w_i \phi(z_i)}{\sum_{i=1}^N w_i} \\
 w_i &= \frac{\tilde{P}(x_{1:n} | z_n) \tilde{P}(z_n)}{Q(z_i | x_{1:n})} \\
 z_i &\sim Q(z_i | x_{1:n})
 \end{aligned}$$

So each particles give an idea of where the person is and we condense this information computing the mean of the positions.

6.3.4 Markov Chain Monte Carlo

The difference with rejection sampling is that each of the x_i are independent sampled. This is what make things time consuming, so the idea is to introduce dependency, the easiest way to do that is Markov model (first order Markov process), so this dependency is obtained through a Markov process.

Given the unnormalized function, the idea is to have a normalized distribution and sample from that, after that we move the distribution and sample again, if we do it many times we'll have a sort of sequence which basically can be modeled as a Markov chain:



How do we generate this distribution? we use our Q that is dependent, it's a conditional distribution function in which what I sample in time $t + 1$ only depends on time t :

$$Q(x^{t+1}|X^t) = Q(x'|x)$$

And how do we generate a Markov chain, we start at time t and sample a new sample dependent on what happens at the previous time:

```

init  $x^0$ 
repeat {
   $t = t + 1$ 
   $x' \sim Q(x^{t+1}|X^t)$  PROPOSAL
   $x^{(t)} \leftarrow x'$  ACCEPTANCE
} until  $t = T$ 

```

Now we can generalize this, the next step is to use an acceptance rule, for instance given the proposal x' we can accept it if the probability of this element to be in the right spot is greater than the probability of the current position I'll accept *if* $\tilde{P}(x') > \tilde{P}(x)$, otherwise I can make a random guess:

```

if  $\tilde{P}(x') > \tilde{P}(x)$ 
  accept the move
else
  make some random guess

```

Metropolis Algorithm

There a simple algorithm, called the METROPOLIS ALGORITHM.

Init, sample from our proposal, then sample some random number u from ad uniform distribution, then if the u is less then the acceptance rule I'll accept the proposal, else I'll reject that means that I'll keep the previous position as the current position and we do this for a certain number of time:

```

Initialize: Set  $x^{(0)}$ ;
repeat
   $t \leftarrow t + 1$ ;
   $x' \sim Q(x' | x^{(t-1)})$ ;
   $u \sim \text{Unif}(0, 1)$ ;
  if  $u < A(x', x^{(t-1)}) = \min \left( 1, \frac{\tilde{P}(x')}{\tilde{P}(x^{(t-1)})} \right)$  then
     $x^{(t)} \leftarrow x'$  ; // ACCEPT
  else
     $x^{(t)} \leftarrow x^{(t-1)}$ ;
  end
until  $t = \tau$ ;

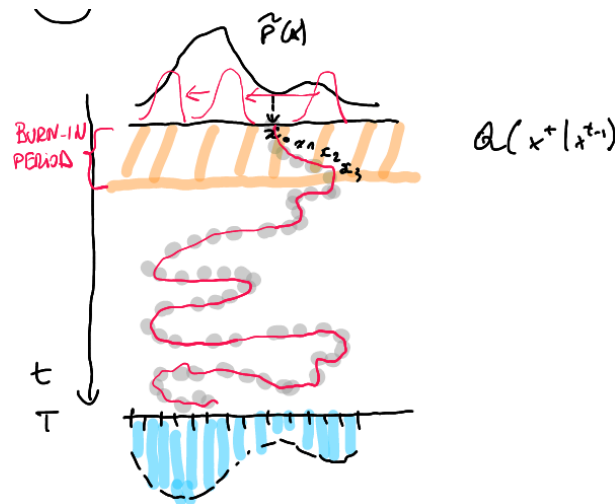
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Algorithm 1: Metropolis-Hastings Algorithm

Given this algorithm an observation that we can do is that this works with the $\tilde{P}(x)$ because this divided by the partition function simply cancel the partition function:

$$\frac{P(x')}{P(x^{(t-1)})} = \frac{\frac{\tilde{P}(x)}{z}}{\frac{\tilde{P}(x^{(t-1)})}{z}}$$

How does it works? we have our distribution and we start sampling, we sampled a lot of points, in the end if we collect all this particles and we project this Markov chain (or random walk) and make an histogram and this histogram prove empirically that we are sampling from the original distribution:



In real case scenario we'll use multiple chain, because this process lead a drift error, in order to mitigate this problem we can introduce a burn period in which we don't consider the sampled particles.

Metropolis Hastings

There is an other algorithm, the METROPOLIS HASTINGS, in which the metropolis part is the acceptance rule, that is exactly the same of the METROPOLIS. The specificity of

the HASTINGS come because the alpha is generalized in this way, the Probability of proposal multiplying two probability sampled from Q:

$$\alpha = \frac{\tilde{P}(x')}{\tilde{P}(x)} \cdot \frac{Q(x|x')}{Q(x'|x)}$$

The METROPOLIS is a specific case because we are assuming that the conditional of sampling a newer point from the current is equal to sampling the current point given the future point, because the fraction will simplified. It's called symmetric proposal:

$$Q(x'|x) \equiv Q(x|x')$$

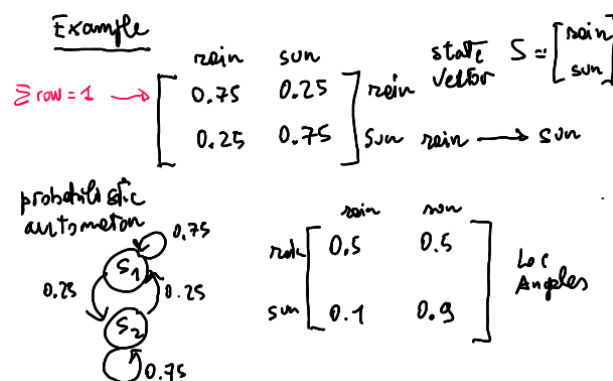
The second question that we can think is WHY does it works. This is the big idea:

1. We use a Markov Chain to sample states of a stationary distribution
2. We designed the Markov Chain such that the stationary distribution is approximately our target distribution stationary distribution $\approx P(x) = \frac{\tilde{P}(x)}{z}$

The idea is that the Markov Chain give us the transition probability:

$$\begin{aligned} P(x^{(t+1)}|x^{(t)}) \\ T(x^{(t)}, x^{(t+1)}) \\ x^t \rightarrow x^{t+1} \end{aligned}$$

In the example we found the probabilistic automaton model, in which we found the probability to stay in the state or change state:



This might change over time, in general we might have this transition probability changing over time (might be time dependent). In general we works with Homogeneous chains, means that for every time the transition will not change:

$$\forall t \ T(x^t, t^{t-1}) = T(x', x)$$

For the next step we might define what is the marginal of the Markov Chain, if we consider t+1 it's obtained of the joint distribution of the x(t+1) and x(t) (we sum over all the possible state at the time t):

$$P(x^{t+1}) = \sum_{x^t} P(x^{t+1}, x^t) = \sum_{x^t} P(x^{t+1}|x^t)P(x^t) = \sum_{x^t} T(x^{t+1}|x^t)P(x^t)$$

We can use this for the definition of invariance (or invariance distribution). We have a specific P^* of x and this is invariant for an homogeneous Markov Chain which we have defined before if P^* of x is equal over the transition probability and P^* . At a certain point I have a distribution I apply my transition mechanism and what happens is that nothing happens we again obtained the original distribution, so the original distribution is like a fixing point. That distribution does not change with the transition machine, it's invariant:

$$P^*(x) = \sum_{x'} T(x', x) P^*(x')$$

This because if it really is the target distribution it might be invariant.

There is an other property, **ERGODICITY**, the idea is to be independently from the initial conditions. My distribution will reach an invariant distribution (equilibrium distribution). Under certain condition equilibrium is equal to invariance:

$$P(x^t) \rightarrow P^*(x) \quad t \rightarrow -\infty$$

It's difficult to satisfy but under certain condition might work.

The last property which is called the **DETAIL BALANCE EQUATION**. We want a P^* distribution is such that satisfied this property.

$$P^*(x) T(x, x') = P^*(x') T(x', x)$$

If this holds we know that P^* is invariant.

The **METROPOLIS HUSTLING CONDITION** satisfies the detail balance equations:

$$\tilde{P}(x^t) Q(x' | x^t) A(x^t, x') = \tilde{P}(x') Q(x^t | x') A(x', x^t)$$

In the end given all such condition what we have is that P^* of x is the stationary distribution that we want. Plus if the Markov Chain is ergodic and irreducible the P^* of x is unique.

6.3.5 Gibbs Sampling



We have a distribution and we want to sample a vector from this distribution:

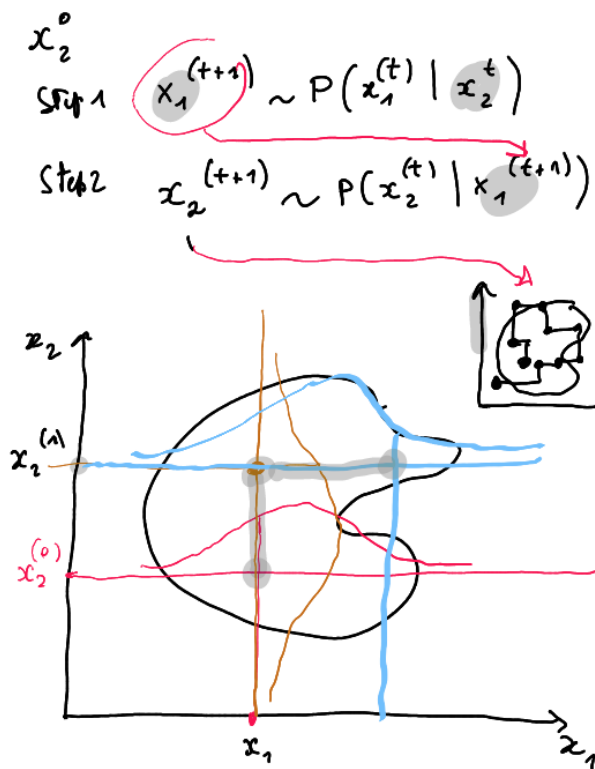
$$x_1, x_2 \sim P(x_1, x_2) \rightarrow x \sim P(x)$$

In order to apply we need some requirements, the requirement is that we have the conditional, that means that we have to exact for of the conditional distribution that gives the probability of x_1 given x_2 $P(x_1|x_2)$ and the probability of x_2 given x_1 $P(x_2|x_1)$. Then the method is very simple, we have an initial condition, assume that we have x_2 at time 0 and for each iteration we start sampling x_1 for the time given x_2 . Once we have x_1 for time $t + 1$ we can sample next x_2 for time $t + 1$ given that we have an x_1 at time $t + 1$ which is the one we have at the previous step and we can go on:

$$x_2^0$$

$$\text{step 1 } x_1^{(t+1)} \sim P(x_1^t | x_2^t)$$

$$\text{step 2 } x_2^{(t+1)} \sim P(x_2^t | x_1^{t+1})$$



So we sample one point at time and then given this point we find a Gaussian distribution and from that we can sample the other point.

Note that Gibbs is a specific case of MM that we always accept. In this term we can rewrite the proposal distribution:

$$Q(x' | x) = P(x'_k | x_{\text{ALL}})$$

And then we can decompose the $P(x)$:

$$P(x) = \tilde{P}(x_k | x_{\text{ALL}}) P(x_{\text{ALL}})$$

$$\alpha = (x', x) = \frac{P(x') Q(x | x')}{P(x) Q(x' | x)} = 1$$

$$A(x', x) = \min\{1, \alpha\}$$

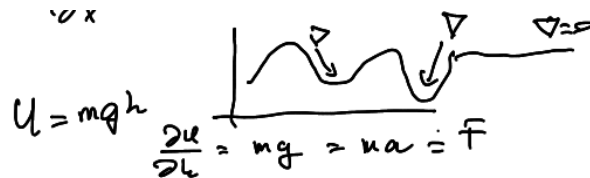
6.4 Langevin Montecarlo

We have our $\tilde{P}(x)$, the unnormalized distribution, and we want to sample x from this distribution, always the same problem, We consider the Langevin Equation:

$$\delta \frac{dx_t}{dt} = -\frac{\partial u(x)}{\partial x} + n(t)$$

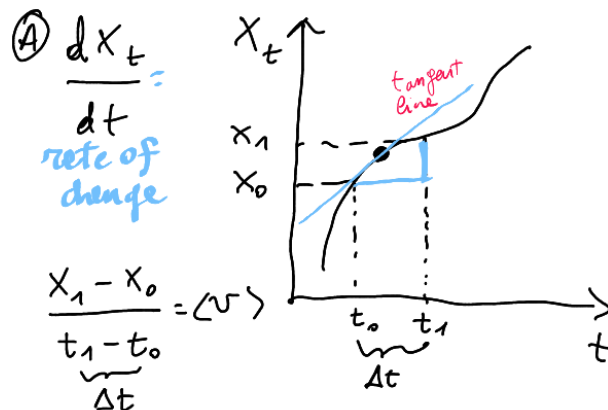
where x_t is the position. So this is speed:

$u = mgh$
 $\frac{\partial u}{\partial h} = mg = ma = F$

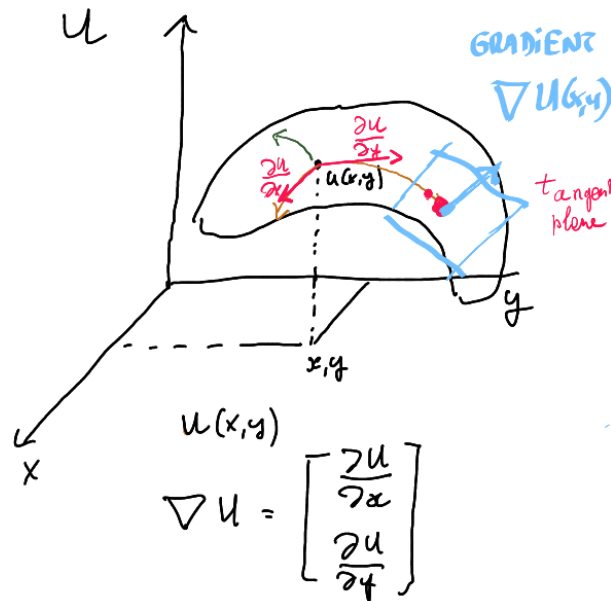


This means that if I consider $\frac{x_1 - x_0}{t_1 - t_0}$ we found the average speed. We can call $t_1 - t_0 = \Delta$ and rewrite the average speed:

$$\lim_{\Delta t \rightarrow 0} \frac{x_1(t_0 + \Delta t) - x_0(t_0)}{\Delta t} = \frac{dx(t)}{dt}$$

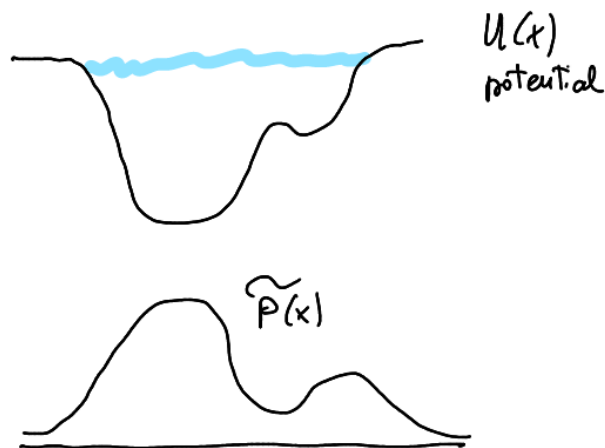


Then we can go to the second part of the equation, $u(x)$ is a potential function, so the second part is a force $-\frac{\partial u(x)}{\partial x} = F$. Imagine we have a function with 2 variables, function u would be some kind of surface in 3 dimension. Now each point on the surface is a point which takes a value (x and y), this is called scalar field in physics. This surface has a variation and we want to measure this variation (that is the speed), but here we have a problem, the direction, because in the previous case we might only follow the line, in this case we might proceed in y or x direction. What we can do is compute at each time the derivative in respect to y and x , this is the partial derivative:



The gradient is a vector which component are the partial derivative of y and x . This gradient act as a force that act as an attractor (goes to the high gradient to the low gradient).

We could imagine this as a profile of a distribution reversing the distribution. If we put a particle in this distribution the particle will stop in the maximum (or minimum) that is what happens in the Maximum Likelihood Estimation.



What can I do to not be stuck in the maximum but explore the distribution is adding the third term of the equation:

$$\frac{dx_t}{dt} = \eta(t) \rightarrow n_t \sim N(o, T^2)$$

It's like we are introducing perturbation.

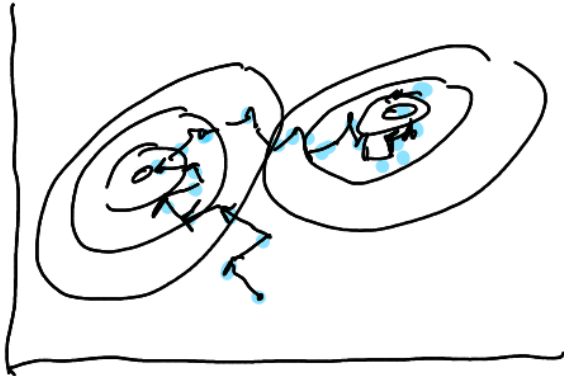
$$\frac{x_{t+1} - x_t}{\tau} = \eta(t)$$

$$x_{t+1} - x_t = \tau \eta(t)$$

$$x_{t+1} = x_t + \tau \eta(t)$$

We sample direction and move following this direction. This is called the drunk walk and technically it's what is called Brownian motion because it's used to describe the random movement of the particles.

Now in the end if we take the full equation and I have, for instance, a potential made by the sum of two Gaussian, I might start sampling randomly and when the particle reach in a Gaussian it's attracted by the gradient. So the Markov chain will be driven by a stochastic component (the noise, noise allow for diffusion, explore) and a deterministic component which potentially drive the chain the high potential zone (allows for the drive):

$$\frac{dX_t}{dt} = - \underbrace{\frac{\partial u(x)}{\partial x}}_{\text{DRIFT}} + \underbrace{\eta(t)}_{\text{noise}} \sim \text{DIFFUSION}$$


6.4.1 Stochastic Differential Equation - SDE

We can write the Langevin Equation in SDE form:

$$\begin{aligned} dX_t &= -\Delta u(x) + \sqrt{2}dB_t \\ \frac{dX}{dt} &= -\frac{\partial u(x)}{\partial x}dt + u(t)dt \\ dB &\implies \frac{dB}{dt} = \eta(t) \end{aligned}$$

We might think of this in a form of differential, in the sense that if we take the derivative of B function it will give the eta. This is a very important equation, what is called in AI diffusion models have this equation as the core, because this is a diffusion equation.

We have two main ideas:

1. The process has a unique invariant form or steady equal state distribution: $x_0 \sim P(x) \implies x_t \sim P(x) t > 0$
2. If we set the $u(x)$ in a later way then we can simulate to sample $x \sim \tilde{P}(x)$

The idea to use Langevin to sampling is basically the same in Montecarlo. Now the basic question is how we take the $u(x)$. Here we have a link to energy based models and the idea is to have $u(x)$ as some kind of energy, for example the log-likelihood :

$$u(x) = E(x)$$

Usually in physics we have models of this kind:

$$P(x) = \frac{e^{-E(x)}}{Z}$$

Where Z is the partition function. This is called GIBBS DISTRIBUTION (describe the equilibrium distribution), obv if we don't have Z we might consider the unnormalized one:

$$\tilde{P}(x) = e^{-E(x)}$$

The connection is by setting exponentiated to the log-likelihood:

$$\begin{aligned} \tilde{P}(x) &= e^{-\log \tilde{P}(x)} \\ \implies u &= -\log \tilde{P}(x) \text{ negative log-likelihood} \end{aligned}$$

Note that we don't need to know $Z(x)$ (the partition function) because we compute the gradient and the partition function is a constant, so the gradient of a constant is equal to 0.

Why this works?

$$dX_t = -\Delta u(x) + \sqrt{2}dB_t$$

We might have multiple Markov Chain and what happens is that we want to describe the collective behavior of many particles. If the dynamics of each particle follow this equation, the corresponding equation is called Fokker-Planck Equation:

$$\frac{\partial P(x, t)}{\partial t} = \frac{\partial}{\partial x} \left[\frac{\partial V}{\partial x} P(x, t) \right] + \frac{\partial^2 P(x, t)}{\partial x^2}$$

This equation is the equation that describe the probability to multiple particles to be in a certain position in which each particle follows the previous equation. In physical therm we might say that the first equation describe the microscopic behavior and the second equation describe the macroscopic behavior.

We are interested in the interval steady-state solution:

$$\frac{\partial P(x, t)}{\partial t} = 0$$

If $P(x) = P_\infty(x, t)$ is solution is:

$$\begin{aligned} \frac{\partial P(x, t)}{\partial t} &= \frac{\partial}{\partial x} \left[\frac{\partial u(x)}{\partial x} P_\infty + \frac{\partial P_\infty}{\partial x} \right] = \frac{\partial}{\partial x} J(x) = 0 \\ \implies J(x) &= \text{constant} \\ \implies J(x) &= \frac{\partial u}{\partial x} P_\infty + \frac{\partial P_\infty}{\partial x} = 0 \\ \implies P(x) &= P_\infty \propto e^{-u(x)} \end{aligned}$$

Simulating the SDE

We have multiple way to discretize the SDE equation. For instance we have the Euler-Maruyana discretization, that relies on this:

$$\frac{dX_t}{dt} \simeq \frac{x_{t+\tau} - x_t}{\tau}$$

In this term we can rewrite the first:

$$\begin{aligned} x_{t+\tau} - x_t &= \tau \Delta \log \tilde{P}(x) + \sqrt{2} dB_t \\ dB_t &= (B_{t+\tau} - B_t) \sim N(0, \tau) \end{aligned}$$

This increments by assumption are independent and Gaussian distributed (have the same Gaussian), so they are i.i.d. increment. In the most simple case in which we have:

$$dX_t = \sqrt{2} dB_t$$

Note that sampling next from $N(0, \tau)$ is equivalent to sample a ξ from a normalized Gaussian and multiplying by standard deviation:

$$x \sim N(0, \tau) = \sqrt{\tau} \xi, \xi \sim N(0, 1)$$

This is the final equation:

$$x_{t+\tau} = x_t + \tau \Delta \log \tilde{P}(x) + \sqrt{\tau} \xi$$

If we have our distribution and sampling, in the end we can make our histogram by the accumulation of the particles that we've sampled. This is called the ULA which means Unadjusted Langevin Algorithm:

Note that the condition is that the gradient must be L-Lipshits (mean that must be strongly continuity).

There is a variant, that is called Metropolis adjusted Langevin (MALA), that is basically the same:

$$x_{t+\tau} \sim N(x_t + \tau \Delta \log \tilde{P}(x),)$$

Now we have a proposal distribution:

$$\begin{aligned} Q(x' | x) &\propto e^{-\frac{\|x' - (x + \tau \Delta \log \tilde{P}(x))\|^2}{2 \cdot 2\tau}} \\ A(x^k, x^{k+1}) &= A(x, x') = \min \left(1, \frac{\tilde{P}(x^{k+1}) Q(x^k | x^{k+1})}{\tilde{P}(x^k) Q(x^{k+1} | x^k)} \right) \end{aligned}$$

6.5 Variational Approximation

This is a kind of deterministic approximation, it's a different kind of approximation in respect to MonteCarlo.

Our objective is to sample a variable from a distribution and then sample other variables given these variables. Under this model comes a lot of modern Machine Learning, starting from Probabilistic PCA (that is continuous problem), Gaussian Mixture Model (basic clustering problem, discrete), Variational Auto-Encoder (VAE) or GAN.

The approach has two key ideas:

1. We use $Q(x)$ to approximate the posterior of a latent variable (like we did in MonteCarlo). Approximating posterior $Q(x) \simeq P(z|x)$ (in MonteCarlo we called it proposal distribution).
2. Use a new measure which is called, the negative free energy $\tilde{\mathbb{F}} = \int_{\mathbb{R}^m} Q(z) \ln \frac{P(x,z)}{Q_z(z)} dz = E_{z \sim Q(z)}[\ln \frac{P(x,z)}{Q_z(z)}]$. Where the number of integral is equal to m. This value is often used both as Evidence Lower Bound (ELB) and as Active Inference Framework (AIF). The free energy function is measuring the error between the prediction and the actual data, so we want to minimize this error.

Where the name free energy come from?

$$\begin{aligned}\tilde{\mathbb{F}} &= \int Q(z) \ln \frac{P(x,z)}{Q(z)} dz \\ &= \int Q(x) \ln P(x,z) - Q(z) \ln Q(z) dz \\ &= E[\ln O(x,z)] - \int Q(z) \ln Q(z) dz\end{aligned}$$

$$\begin{aligned}E[\ln O(x,z)] &\implies \langle \ln P(x,z) \rangle \text{ average internal energy } u \quad \int Q(z) \ln Q(z) dz \implies \text{Entropy } S \\ &= u - S\end{aligned}$$

In physics free energy is defined as $F = u - TS$ where T is the temperature in this case we can set the temperature as 1 and from this is where derive the idea of free energy. Since we have a family of distribution what we obtained is that the best distribution to choose, the one that better approximate the original distribution, is the one that maximize the free energy.

In order to arrive to this energy we need some background. The first tool is the Jensen inequality:

$$E[f(x)] \geq f(E[x])$$

Imagine that we have convex curve and we have points, the center of these point will be empirically the mean of the points:

The second tool is the Kullback-Liblen divergence:

$$\begin{aligned}KL(Q||P) &= \int Q(z) \ln \frac{Q(z)}{P(z)} dz \\ &= E_{Q(z)}[\ln \frac{Q(z)}{P(z)}]\end{aligned}$$

Imagine that we have two distribution, the divergence is a sort of difference between the two distributions:

$$\begin{aligned}KL(Q_1||P) &> KL(Q_2||P) \\ KL(Q||P) &= 0 \Leftrightarrow Q = P \\ KL(Q||P) &\geq 0\end{aligned}$$

When this measure is equal to 0 means that the two distribution are the same. We know that in general the KL between two distribution will be greater than 0.

So theoretically to find the optimal Q:

$$Q^* = \arg \min_Q KL(Q||P)$$

So with some techniques (like gradient descent) we try to shrink the divergence between the two distributions.

The third tool is Euler-Lagrange equation. Assume we have a path called $x(t)$, and between two points for sure will be a minimal path (in Euclidean geometry assumption this path will be a line).

We define a functional over the path, for instance I might define first functional as the integral between the two point of the square norm of t:

$$F_1[x] = \int_{t_1}^{t_2} |x(t)|^2 dt$$

If we minimize this function we'll have the minimum length path. We might derive a second function, which is derived from the integral between two point, taken used the physicist notation x dot, that stands for the time derivative of $x(t)$:

$$F_2[x] = \int_{x_1}^{x^2} |\dot{x}(t)|^2 dt$$

Minimizing this function means finding the minimal energy path. The problem here is find the minimal path, so if I have this path and some variation:

$\epsilon(t)$ is called the variation of perturbation over the minimal path. We define a functional:

$$\begin{aligned} F[\tilde{x}] &\leq F[x] \\ \delta f &= F[x] - F[\tilde{x}] \iff \text{variation of the functional} \end{aligned}$$

What happens here is that the shortest path $\tilde{x}$ minimize δf , and $\delta f = 0$ is the Euler-Lagrange Equations where $F[x] = \int_{t_1}^{t_2} f(x, \dot{x}, t) dt$ (x position, $\dot{x}$ velocity, t time).

$$\begin{aligned} F[\tilde{x}] &\leq F[x] \\ \int_{t_1}^{t_2} f(\tilde{x} + \epsilon, \dot{\tilde{x}} + \dot{\epsilon}, t) \end{aligned}$$

We are going to do Taylor Expansion of the first order:

$$\begin{aligned}
&\approx \int_{t_1}^{t_2} [f(\tilde{x}, \dot{\tilde{x}}, t) + \frac{\partial f}{\partial x} \epsilon + \frac{\partial f}{\partial \dot{x}} \dot{\epsilon}] dt \\
&\int_{t_1}^{t_2} f(\tilde{x}, \dot{\tilde{x}}, t) + \int_{t_1}^{t_2} \frac{\partial f}{\partial x} \epsilon dt + \int_{t_1}^{t_2} \frac{\partial f}{\partial \dot{x}} \dot{\epsilon} dt \\
&\int_{t_1}^{t_2} f(\tilde{x}, \dot{\tilde{x}}, t) = F[x] \\
\mathbf{A} &= \int_{t_1}^{t_2} \frac{\partial f}{\partial \dot{x}} dt = \text{integration by parts} \\
&= [\frac{\partial f}{\partial \dot{x}} \epsilon(t)]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} (\frac{\partial f}{\partial \dot{x}} \epsilon) dt \\
F[\tilde{x}] &\leq F[x] + \int_{t_1}^{t_2} \frac{\partial f}{\partial x} \epsilon dt - \int_{t_1}^{t_2} \frac{d}{dt} (\frac{\partial f}{\partial \dot{x}} \epsilon) dt \\
F[\tilde{x}] + \int_{t_1}^{t_2} \frac{\partial f}{\partial x} \epsilon dt - \int_{t_1}^{t_2} \frac{d}{dt} (\frac{\partial f}{\partial \dot{x}} \epsilon) dt &\Leftarrow F[x] \\
F[x] - F[\tilde{x}] &= \int_{t_1}^{t_2} [\frac{\partial f}{\partial x} - \frac{d}{dt} (\frac{\partial f}{\partial \dot{x}})] \epsilon dt \implies F[x] - F[\tilde{x}] = \delta f[x] \text{ variation} \\
\delta F = 0 &\implies \frac{\partial f}{\partial x} - \frac{d}{dt} (\frac{\partial f}{\partial \dot{x}}) = 0 \implies \text{Euler Lagrange Equation}
\end{aligned}$$

To given a physical example:

$$\begin{aligned}
F(x, \dot{x}, t) &= L(x, \dot{x}, t) \\
&= T - u(x) = \frac{1}{2} m \dot{x}^2 - u(x) \\
\frac{\partial L}{\partial x} &= -\frac{\partial u}{\partial x} \\
\frac{\partial L}{\partial \dot{x}} &= \frac{\partial}{\partial \dot{x}} \left(\frac{1}{2} m \dot{x}^2 \right) = m \dot{x} \\
\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{x}} \right) &= m \ddot{x} \\
&\implies -\frac{\partial u(x)}{\partial x} = m \ddot{x} \\
F &= m \ddot{x} = ma
\end{aligned}$$

6.5.1 Free energy theorem

It lives under the assumption that the Qz taken in the mean field approximation, the optimal Q_z^* which approximates the free energy $\tilde{F}[Q(z)]$ in